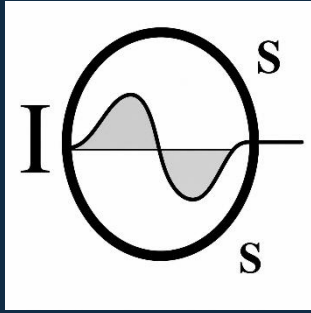




INSPECTION & SPECIFICATION SERVICES CC.

CONTROLLED ENGINEERING INSPECTION REPORT

CABLE BELT LOWER ANGLE STATION SHEAVES

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MECHANICAL ASSET CONDITION ASSESSMENT

Cable Belt Lower Angle Station Sheaves Inspection Report

Inspection report · engineering assessment · maintenance recommendations



Client	South32	Mine / site	Wessels Mine
Area / system	Cable Belt - Lower Angle Station	Asset / asset group	Lower Angle Station Sheaves 1-4
Report number	ISS_IRS32-Wessels-200726	Revision	1.1
Issue date	17/08/2026	Inspection date	20 July 2026
Issue status	Approved	Prepared by	Brendan Wood
Reviewed by	Kevin Wood	Approved by	Kevin Wood
Client reference	South32 Wessels	ISS job number	IRS32-Wessels-200726

CONFIDENTIALITY & RELIANCE

Prepared for the named client and stated purpose only. Reliance, distribution and operational decisions remain subject to the agreed scope, contractual requirements, stated limitations and responsible engineering review.

Controlled Document Notice

This controlled report records the Cable Belt Lower Angle Station sheave inspection data, photographic evidence, engineering reference drawings, engineering assessment and recommendations for South32 Wessels Mine. Information not available at the time of issue is identified as “Not supplied” and remains subject to the limitations and follow-up requirements stated in this report.

The report findings are limited to the agreed scope, access conditions, inspection methods, information supplied and evidence recorded. Visual inspection does not establish the internal condition of bearings, shafts, welds or other inaccessible components unless supporting examination or test evidence is recorded.

Only external standards, acceptance criteria, measurements and NDT results verified for this project are referenced in this report.

Document Control

Table 1 - Document control and report metadata

Report title	Cable Belt Lower Angle Station Sheaves Inspection Report
Report number	ISS_IRS32-Wessels-200726
Revision	1.1
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Area / system	Cable Belt - Lower Angle Station
Asset / asset group	Lower Angle Station Sheaves 1-4
Template version	2.1

Document Authorization Record

Table 2 - Document authorization record

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Approved by	Kevin Wood	Managing Engineer	iss@inspecserv.co.za	17/08/2026	KC

Revision History

Table 3 - Revision history

Revision	Date	Issue status	Description of change	Prepared by	Approved by
0.1	02/08/2026	Draft	Lower Angle Station findings, reference drawings and supplied photographic evidence populated	Brendan Wood	Brendan Wood
0.5	08/08/2026	Draft	Preliminary compilation from inspection notes and supplied photographic evidence.	Brendan Wood	Brendan Wood
1.1	17/08/2026	Approved	Final engineering assessment, system-level alignment recommendations and report issue review completed	Brendan Wood	Kevin Wood

Distribution List

Table 4 - Controlled distribution list

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1 Purpose

This report documents the condition assessment of the four Cable Belt Lower Angle Station sheaves at South32 Wessels Mine. It records the inspection observations, groove-depth readings, asymmetric groove-wear patterns, rope-path/fleet-angle geometry, engineering interpretation and planned corrective actions. Based on the physically observed angular rope path, the common asymmetric wear pattern identified on all four Lower Angle Station sheaves and the supporting system evidence, the non-zero fleet-angle/off-centre rope-tracking condition is assessed as the principal probable cause of the recurring asymmetric groove wear. Quantitative geometry measurements are still required to define the magnitude of the offsets and to finalize the corrective design. The engineering objective is to reduce accelerated wear of the 45 mm cable-belt rope, restore centred rope support through the sheave grooves, limit abnormal load transfer into the bearing systems and improve the service life of the corresponding Upper and Lower Angle Station sheaves.

2 Executive Summary

Four Cable Belt Lower Angle Station sheaves are included in this assessment. All four were physically inspected, measured and found to exhibit the same general asymmetric groove-contact/wear condition, although usable groove photographs are available only for Sheaves 2 and 3 because of a camera technical problem encountered during the inspection. All four are recorded as Poor condition, Level 2 - Moderate severity and P3 - Planned maintenance priority. Recorded groove-depth readings are 104 mm, 106 mm, 105 mm and 102 mm for Sheaves 1 to 4 respectively. The supplied OEM plate identifies 97 mm as the nominal rope-groove depth and 127 mm as the OEM-defined maximum wear depth, giving a 30 mm vertical wear range. The current readings therefore remain within the supplied vertical wear limit; however, the dominant concern is the asymmetric lateral sidewall wear, because the groove is no longer providing optimum centred support to the rope and may locally compress, pinch or overstress rope strands. During the inspection, the bearing temperature-monitoring sensors on Lower Sheaves 3 and 4 were also found not attached to their respective bearings and require reattachment and functional verification.

South32 Wessels Mine has advised that the 45 mm cable-belt rope was replaced approximately 20 to 24 months ago and has already experienced wear approaching 50% of its allowable wear allowance, compared with an expected minimum service life of approximately 6 to 8 years. The rope wear condition has also been assessed by an independent inspection contractor; that contractor's formal report was not included in the present report package. Exact replacement dates, historical wear values and the reported 12-to-18-month Angle Station sheave service trend remain mine-controlled information and should be verified directly with the mine Condition Monitoring / maintenance functions where confirmation is required. Release of the independent contractor report remains at the mine's discretion. A catastrophic failure of Upper Angle Station Sheave No. 4 occurred on 14 July 2026, with bearing collapse identified as the failure mode. Appendix F documents the collapsed/displaced bearing, metallic debris, post-extraction scoring/smearing and asymmetric groove/contact condition. This evidence corroborates the wider system concern but does not, by itself, constitute a complete root-cause analysis of that individual failure.

The physically observed rope path transitions through a pronounced angular offset between the corresponding Upper and Lower Angle Station sheaves. At the Lower station this produces a non-zero entry fleet angle and off-centre groove contact; the same geometry also governs rope departure and the corresponding Upper station interfaces. On the balance of the observed geometry and wear evidence, this condition is assessed as the principal probable cause of the recurring asymmetric wear. Corrective action should therefore address the rope path rather than repeatedly replace damaged components while leaving the geometry unchanged. The recommended corrective solution, subject to quantitative survey and detailed design verification, is engineered guide control at both the entry and departure of the affected Upper and Lower Angle Station main sheaves so that the rope enters and leaves each main sheave on its intended groove centre plane at 0 degrees fleet angle, or as close as reasonably practicable. The conceptual drawings in this report illustrate the functional intent only and do not prescribe the final quantity, position or configuration of guide sheaves. Existing worn Lower Angle Station main sheaves are to be replaced with new sheaves when the corresponding guide-sheave correction is commissioned, unless earlier replacement is required by condition-monitoring or other deterioration.

2.1 Executive Dashboard

ASSETS INSPECTED 4	OVERALL CONDITION Poor	HIGHEST PRIORITY P3	OPEN ACTIONS 6
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Table 5 - Executive asset summary matrix

Asset	Asset tag / ID	Overall condition	Highest severity	Priority	Open findings	Executive comment
Cable Belt Lower Angle Station Sheave 1	Not supplied	Poor	Level 2	P3	1	Groove depth 104 mm; asymmetric groove-side wear directly observed and recorded. No usable groove photographs due to a camera technical issue.
Cable Belt Lower Angle Station Sheave 2	Not supplied	Poor	Level 2	P3	1	Groove depth 106 mm; front-view photographs show asymmetric sidewall/groove-base wear. Wear appears left from the front and right from the rear.
Cable Belt Lower Angle Station Sheave 3	Not supplied	Poor	Level 2	P3	1	Groove depth 105 mm; front-view photographs show asymmetric sidewall/groove-base wear. Wear appears left from the front and right from the rear. Bearing temperature sensor found not attached; rectify and verify.
Cable Belt Lower Angle Station Sheave 4	Not supplied	Poor	Level 2	P3	1	Groove depth 102 mm; asymmetric groove-side wear directly observed and recorded. OEM plate and groove-limit photographs supplied. Bearing temperature sensor found not attached; rectify and verify.

Table 6 - Executive recommendation register

Action ID	Asset	Recommended action	Priority	Responsible discipline	Target / shutdown	Status
REC-001	Complete Angle Station rope path - Lower and Upper	Complete quantitative rope-path / alignment / fleet-angle survey and the outstanding Upper Sheaves 1-3 condition inspection during the design-discovery phase	P3	Mechanical / Engineering	Short term / before design finalization	Open
REC-002	Complete Angle Station rope path - Lower and Upper	Develop the recommended entry-and-departure guide-sheave correction; final quantity/configuration to be determined by specialist detailed design	P3	Mechanical / Engineering	Design short term; install at approved planned shutdown	Open
REC-003	Lower Angle Station Sheaves / bearings	Restore Sheaves 3 and 4 bearing temperature sensors and continue enhanced mine condition monitoring; escalate any abnormal bearing/rope/groove trend	P3	Mechanical / Condition Monitoring	Immediate rectification / ongoing until correction	Open

Action ID	Asset	Recommended action	Priority	Responsible discipline	Target / shutdown	Status
REC-004	Lower Angle Station main sheaves and corresponding Upper positions	Replace each existing worn Lower main sheave with a new sheave when its guide-sheave correction is installed; replace earlier if condition monitoring indicates deterioration	P3	Mechanical / Maintenance	At modification shutdown or earlier if condition requires	Open
REC-005	Modified Angle Station rope paths	Verify as-left geometry and centred dynamic rope tracking, then trend groove wear, rope wear and bearing vibration/temperature against the pre-correction baseline	P3	Mechanical / Inspection / Condition Monitoring	Commissioning and 6–12-month follow-up	Open
REC-006	Angle Station / Cable Belt system	Record/confirm exact rope replacement, wear and sheave service history from mine-controlled records; reference independent rope-inspection evidence where released	P3	Reliability / Maintenance Planning	Short term / baseline establishment	Open

3 Introduction

Inspection & Specification Services cc. was requested to document the condition of the Cable Belt Lower Angle Station sheaves at South32 Wessels Mine. The Angle Station comprises four corresponding Upper Angle Station sheaves and four Lower Angle Station sheaves. The current report formally assesses the four Lower Angle Station sheaves. Detailed inspection of the remaining Upper Angle Station sheaves was intended as part of the inspection opportunity, but the plant was required to return to production before that work could be completed. Upper Angle Station Sheave No. 4 failure evidence is included only as supporting system context; no complete condition classification is assigned to the other Upper sheaves in this report.

The angular transition of the rope between the Upper and Lower Angle Station sheaves was physically observed during the site inspection and is also evident from the system layout. The rope path is not coincident with the respective main-sheave groove centre planes. Consequently, the rope approaches and/or departs the Angle Station sheaves at non-zero fleet angles and preferentially loads one side of the groove. The same general asymmetric wear pattern was physically identified on all four Lower Angle Station sheaves. The non-zero fleet-angle/off-centre tracking mechanism is therefore assessed as the principal probable cause of the recurring lower-sheave sidewall wear, while a quantitative survey is still required to measure the actual angles and offsets and to define the final corrective geometry.

The assessment is based on the inspection notes, groove-depth readings, direct visual observations, available Lower Angle Station photographs, engineering reference drawings and the supplied Upper Angle Station Sheave No. 4 failure photographs. The report inspection date is 20 July 2026. Photographic coverage is incomplete for Lower Sheaves 1 and 4 because of a camera technical problem, but their asymmetric groove condition was directly observed and recorded during the inspection. Quantitative fleet-angle/alignment readings were not taken. The available rope-wear condition has been independently inspected by another contractor; its formal report is not included here, and exact historical dates/service intervals remain for the mine to confirm from Condition Monitoring and maintenance records.

4 Scope of Inspection

Table 7 - Assets and activities included in scope

Asset	Asset tag / ID	Location	Inspection activity	Access / operating state	Comment
Lower Angle Station Sheave 1	Not supplied	Cable Belt Lower Angle Station	Visual inspection / groove-depth measurement / rope tracking observation	Stopped / isolated state	Included in report scope
Lower Angle Station Sheave 2	Not supplied	Cable Belt Lower Angle Station	Visual inspection / groove-depth measurement / rope tracking observation	Stopped / isolated state	Included in report scope
Lower Angle Station Sheave 3	Not supplied	Cable Belt Lower Angle Station	Visual inspection / groove-depth measurement / rope tracking observation	Stopped / isolated state	Included in report scope
Lower Angle Station Sheave 4	Not supplied	Cable Belt Lower Angle Station	Visual inspection / groove-depth measurement / rope tracking observation	Stopped / isolated state	Included in report scope

Table 8 - Scope exclusions

Excluded item	Reason / access constraint	Assessment implication	Recommended follow-up
Upper Angle Station Sheaves 1-3 detailed condition assessment	Detailed inspection was not completed because the plant was required to return to production before the remaining Upper Angle Station sheaves could be inspected.	No complete defect classification is assigned to Upper Sheaves 1-3. The angular rope transition between the Upper and Lower stations was physically observed, and the condition of the uninspected Upper sheaves is inferred only to the extent stated in the engineering assessment.	Complete the outstanding Upper Angle Station inspection during the quantitative survey / design-discovery phase, and include all four Upper sheaves in the final corrective design.

5 Inspection Limitations

The inspection was limited by the recorded access, operating condition, visibility, cleanliness, disassembly status, information supplied and inspection methods. Internal bearing condition must not be inferred from external visual inspection unless supporting evidence is available.

NDT results may be reported only where the method, scope, operator or report reference and result evidence are available.

Table 9 - Inspection limitations register

Limitation ID	Affected asset / area	Limitation	Impact on assessment	Follow-up requirement
LIM-001	All Lower Angle Station Sheaves	The angular rope transition/non-zero fleet-angle condition was physically observed, but quantitative fleet-angle and lateral-offset measurements were not recorded.	The existence and direction of the angular rope path can be assessed qualitatively, but its exact magnitude and the final guide-sheave geometry cannot be calculated from this inspection alone.	Perform a quantitative rope-path/alignment survey before final corrective design.
LIM-002	Lower Sheaves 1 and 4	Usable groove photographs are not available because of a camera technical problem encountered during the inspection. Both sheaves were nevertheless visually inspected, measured and their	Photographic traceability is weaker than for Sheaves 2 and 3, but the finding is based on direct inspection observation rather than inference alone.	Retain the recorded field observations and obtain comparative photographs at the next inspection.

Limitation ID	Affected asset / area	Limitation	Impact on assessment	Follow-up requirement
		asymmetric wear conditions were recorded in the field notes.		
LIM-003	Bearings - all Lower Angle Station sheaves	Internal bearing condition, bearing geometry, rope tension and vibration/temperature data were not included in the supplied evidence.	The report can describe the probable load-path effect of off-centre rope loading but cannot calculate bearing reactions or assign internal bearing condition.	Request available bearing arrangement and condition-monitoring information from the mine Condition Monitoring department.
LIM-004	Photographic left/right orientation	Left/right descriptions are viewpoint-dependent. Sheave 2 and Sheave 3 wear photographs used in this report are Front View images. The same physical wear location appears on the opposite left/right side when viewed from the Rear.	Without a stated viewing direction, side-specific descriptions can appear contradictory even though the physical wear location has not changed.	State Front View or Rear View on all side-specific photographic evidence and interpret left/right using CBASS-REF004.
LIM-005	Upper Angle Station Sheaves 1-3	The planned detailed inspection of the remaining three Upper Angle Station sheaves could not be completed because the plant was returned to production.	The angular rope path between the Upper and Lower stations was observed, but detailed groove/bearing condition on Upper Sheaves 1-3 is not established by this report.	Complete the outstanding Upper Angle Station Sheaves 1-3 condition inspection during the survey / design-discovery phase.
LIM-006	Cable Belt rope and Angle Station service history	The rope wear condition has been independently inspected by another contractor, but that contractor's formal report is not included in this package. Exact rope replacement dates, wear percentage at specific dates and historical Angle Station sheave replacement intervals remain mine-record information.	The reported history is relevant system context, but exact values should not be treated as ISS-derived life statistics unless the supporting mine/third-party records are incorporated.	Where confirmation is required, verify exact values directly with the mine Condition Monitoring / maintenance functions. The independent rope-inspection report may be released at the mine's discretion.

6 Applicable Standards and References

Table 10 - Applicable standards and references register

Ref. ID	Type	Title / description	Document number	Revision / date	Source	Status
REF-001	Engineering reference drawing	Groove wear profile context - CBASS-REF001 to CBASS-REF003	CBASS-REF001 to 003	Rev 1.0	Supplied report package	Reference only
REF-002	Engineering reference drawing	Front-Rear orientation and rope-path geometry - CBASS-REF004 to CBASS-REF008	CBASS-REF004 to 008	Rev 1.0	Supplied report package	Reference only
REF-003	Engineering reference drawing	Angular forces on sheave bearings	CBASS-REF009	Rev 1.0	Supplied report package	Reference only
REF-004	OEM / asset plate	Kimberley Engineering Works sheave attributes and groove-depth limit plate (Sheave 4 photographs)	CBASSL4-REF001 / REF002	Photo evidence	Supplied report package	Photographic evidence

Ref. ID	Type	Title / description	Document number	Revision / date	Source	Status
REF-005	Supporting photographic evidence	Upper Angle Station Sheave No. 4 failure / post-extraction photographs	CBASSUF4-REF001 to REF014; CBASSU4-REF001 to REF005	July 2026 photographs	Supplied report package	Contextual reference only

7 Definitions and Abbreviations

Table 11 - Definitions and abbreviations

Term	Definition
IR	Inspection Report
ISS	Inspection & Specification Services cc.
NDT	Non-destructive testing
UT	Ultrasonic testing
MPI	Magnetic particle inspection
OEM	Original equipment manufacturer
P1-P4	Maintenance priority classifications defined in this report
Fleet angle	Angular deviation of the rope path from the sheave groove/centreline at entry or exit; a non-zero angle can produce sidewall contact.
Front View	View from the Cable Belt/front side of the Lower Angle Station sheaves looking toward the rear. A side-specific wear feature must be described relative to this stated viewpoint.
Rear View	View from the opposite/rear side looking toward the Cable Belt/front. A physical feature viewed from 180 degrees opposite will appear on the opposite left/right side without having changed location.
Principal probable cause	The engineering mechanism assessed, on the balance of the observed geometry and condition evidence, as the most likely cause of the recurring condition. Quantitative survey or dedicated failure analysis may still be required to measure the mechanism or prove sole causation for an individual failure.

8 Responsibilities

Table 12 - Responsibility and interface matrix

Party / role	Responsibility	Report interface / deliverable
Client management / responsible site authority	Provide scope confirmation, access, isolation, asset information, operational context and formal site decisions.	Review findings, determine operational response and authorize corrective actions.
ISS inspector / project manager	Perform the agreed inspection, record evidence, compile findings and recommendations within the stated limitations.	Prepare and issue the controlled inspection report.
ISS reviewing engineer	Review technical content, evidence traceability, engineering assessment and recommendations.	Technical review and approval.
Maintenance contractor / specialist, where appointed	Verify repair scope, execute approved work and provide completion evidence.	Repair records, measurements, NDT and close-out evidence.

SECTION B

ASSET & INSPECTION
FRAMEWORK

Asset data, inspection equipment, methodology, assessment criteria and reference orientation.



9 Asset Information

Four lower angle-station sheaves are included. Unknown values are identified as Not supplied.

9.1 Lower Angle Station Sheave 1

Asset name	Cable Belt Angle Station Lower Sheave 1	Asset tag / ID	Not supplied
Location	Cable Belt Lower Angle Station	Service / duty	Cable belt rope redirection at lower angle station
Manufacturer / OEM	Kimberley Engineering Works (Pty) Ltd.	Model / serial number	Not supplied
Sheave diameter	1900 mm	Number of grooves	Not supplied
Rope diameter	45 mm	Groove profile / radius	Not supplied
Bearing type	Not supplied	Shaft / axle reference	Not supplied
Design speed / duty	Not supplied	Commissioned / installation date	Not supplied
Drawing / OEM reference	Not supplied	Inspection date	20 July 2026
Inspector(s)	ISS inspection team	Overall condition rating	Poor

9.2 Lower Angle Station Sheave 2

Asset name	Cable Belt Angle Station Lower Sheave 2	Asset tag / ID	Not supplied
Location	Cable Belt Lower Angle Station	Service / duty	Cable belt rope redirection at lower angle station
Manufacturer / OEM	Kimberley Engineering Works (Pty) Ltd.	Model / serial number	Not supplied
Sheave diameter	1900 mm	Number of grooves	Not supplied
Rope diameter	45 mm	Groove profile / radius	Not supplied
Bearing type	Not supplied	Shaft / axle reference	Not supplied
Design speed / duty	Not supplied	Commissioned / installation date	Not supplied
Drawing / OEM reference	Not supplied	Inspection date	20 July 2026
Inspector(s)	ISS inspection team	Overall condition rating	Poor

9.3 Lower Angle Station Sheave 3

Asset name	Cable Belt Angle Station Lower Sheave 3	Asset tag / ID	Not supplied
Location	Cable Belt Lower Angle Station	Service / duty	Cable belt rope redirection at lower angle station
Manufacturer / OEM	Kimberley Engineering Works (Pty) Ltd.	Model / serial number	Not supplied
Sheave diameter	1900 mm	Number of grooves	1
Rope diameter	45 mm	Groove profile / radius	Not supplied
Bearing type	Not supplied	Shaft / axle reference	Not supplied
Design speed / duty	Not supplied	Commissioned / installation date	Not supplied
Drawing / OEM reference	Not supplied	Inspection date	20 July 2026
Inspector(s)	ISS inspection team	Overall condition rating	Poor

9.4 Lower Angle Station Sheave 4

Asset name	Cable Belt Angle Station Lower Sheave 4	Asset tag / ID	Not supplied
Location	Cable Belt Lower Angle Station	Service / duty	Cable belt rope redirection at lower angle station
Manufacturer / OEM	Kimberley Engineering Works (Pty) Ltd.	Model / serial number	Job No. 6664
Sheave diameter	1900 mm	Number of grooves	1
Rope diameter	45 mm	Groove profile / radius	Not supplied
Bearing type	Not supplied	Shaft / axle reference	Not supplied
Design speed / duty	Not supplied	Commissioned / installation date	01-2022
Drawing / OEM reference	CBASSL4-REF001 / CBASSL4-REF002	Inspection date	20 July 2026
Inspector(s)	ISS inspection team	Overall condition rating	Poor

10 Inspection Team and Equipment Used

10.1 Inspection Team

Name	Role / discipline	Company	Competency / authorization	Site induction / permit reference
Brendan Wood	Inspector / Projects and Systems	ISS	Technician	N/A
Kevin Wood	Technical reviewer / Managing Engineer	ISS	Managing Engineer	N/A

10.2 Inspection Equipment

Equipment is recorded only to the extent supported by the ISS inspection records. Calibration and manufacturer accuracy information was not independently verified in the supplied package.

Equipment	Purpose	Manufacturer / model	Equipment ID	Calibration certificate	Calibration due	Accuracy / range	Comment
Digital camera / mobile device	Photographic documentation	Not recorded	Not recorded	N/A	N/A	N/A	Used for visual evidence record.
Laser distance meter	Groove-depth reading	Bosch PLR 50	009741746	Not supplied	Not verified	Manufacturer accuracy/range not verified from supplied documentation	Used in the sheave groove-depth measurements.
Steel rule	Contextual indication of rope-path offset	Not recorded	Not recorded	Not recorded	Not recorded	Not recorded	Photograph provides contextual scale only; no formal alignment value recorded.

11 Inspection Methodology

The inspection comprised visual examination, photographic documentation, groove-depth measurement, rope-groove assessment and qualitative rope-tracking/alignment observation. External structural, fastener, corrosion/coating and bearing-housing conditions were assessed only where visible and accessible. No NDT results were included in the supplied inspection evidence.

Table 13 - Inspection methodology matrix

Method ID	Inspection activity	Purpose	Evidence captured	Acceptance / reference basis	Limitation / note
M-01	Visual inspection	Record visible condition, damage, wear and anomalies	Observation notes and photographs	Verified reference or engineering assessment	Limited to visible/external condition recorded in supplied photographs and notes.
M-02	Rope groove assessment	Assess visible groove profile, contact and wear indicators	Photographs / gauge / measurement	OEM / verified criterion / comparative evidence	Groove-depth readings supplied; equipment calibration certificate not supplied.
M-03	Rope tracking and alignment observation	Record visible tracking indicators and abnormal contact evidence	Photographs / alignment readings	Verified drawing / measurement basis	Qualitative only; no formal fleet-angle/alignment readings supplied.

12 Inspection Criteria

Table 14 - Inspection criteria and evidence requirements

Attribute	Assessment basis	Evidence required	Typical record	Reference / note
Rope groove and rim condition	Visible wear, damage, profile concerns and abnormal contact evidence	Photographs and measurements where performed	Finding / defect ID	OEM groove-depth plate CBASSL4-REF002; Figures 1–3; Sections 16.1–16.4
Rope tracking and alignment	Visible tracking evidence, contact pattern, rope approach and departure	Photographs, alignment observations and readings where performed	Finding / defect ID	CBASS-REF004 to CBASS-REF008; Sections 16.1–16.5
Hub, shaft and bearing housing	Visible external condition, security, leakage and damage indicators	Photographs / notes / temperature or vibration data where included	Observation ID	CBASS-REF009; OBS-005 and OBS-006; Section 16.5; Appendix F for Upper Sheave No. 4 contextual evidence
Structural components and supports	Visible deformation, cracking indicators, corrosion, coating and fastener condition	Photographs / NDT / measurements where included	Finding / defect ID	Visual inspection only; no dedicated structural defect or NDT result assigned within this inspection scope

13 Condition Rating, Defect Severity and Maintenance Priority

The following classifications support consistent reporting and maintenance planning. They do not replace the mine's formal risk assessment, statutory obligations or decisions by the responsible site authority.

Table 15 - Condition rating

Rating	General meaning	Typical reporting response
Excellent	No material defect observed within the inspection scope.	Record condition and routine inspection interval.
Good	Minor condition observations only.	Monitor through routine maintenance and inspection.
Fair	Serviceable with deterioration requiring planned attention.	Plan maintenance, measurement or enhanced monitoring.
Poor	Significant deterioration requiring priority engineering or maintenance review.	Define corrective action and verification requirements.
Critical	Potentially unacceptable condition requiring immediate responsible-site review.	Escalate immediately and determine operational response.

Table 16 - Defect severity classification

Level	Classification	Definition	Default action
Level 1	Minor	Low-significance observation requiring monitoring or routine correction.	Record and monitor.
Level 2	Moderate	Defect requiring planned corrective action.	Schedule corrective work.
Level 3	Significant	Defect requiring priority engineering review and corrective action.	Repair at the next suitable shutdown or earlier.
Level 4	Critical	Potentially critical condition requiring immediate responsible-site review.	Immediate escalation and corrective action.

Table 17 - Maintenance priority guidance

Priority	Meaning	Typical timeframe guidance	Required response
P1	Immediate	Before continued operation where required by responsible review.	Escalate and define immediate controls / corrective action.
P2	Urgent	Earliest practicable maintenance opportunity.	Plan and execute short-term corrective action.
P3	Planned	Next suitable scheduled shutdown or maintenance cycle.	Include in approved maintenance plan.
P4	Monitor	Routine inspection and maintenance interval.	Trend and verify at future inspection.

13.1 Application of Condition Rating, Severity and Priority to the Lower Angle Station Sheaves

The supplied OEM plate identifies 97 mm as the nominal rope-groove depth and 127 mm as the OEM-defined maximum wear depth. The 30 mm difference is treated in this report as the available vertical groove-depth wear range. The recorded depths of 102-106 mm have therefore consumed approximately 5-9 mm of that vertical range and remain below the 127 mm maximum wear-depth limit. Reaching or exceeding 127 mm is to be treated as an out-of-service / retirement condition because it is the maximum wear depth defined on the OEM specification plate.

The Poor / Level 2 - Moderate / P3 - Planned classification is driven by the functional condition of the groove, not by vertical depth alone. The asymmetric sidewall wear means the groove is no longer providing optimum centred support to the 45 mm rope and can create localized strand compression, pinching, contact stress and fatigue, with potential accelerated wear of both rope and sheave. The condition is therefore rated Poor. It remains Level 2 - Moderate because the recorded vertical wear limit has not been exceeded and the available evidence does not require immediate component retirement solely on groove depth; however, planned corrective action is required. P3 - Planned provides a balanced maintenance timeframe in which the mine can, subject to its operational risk controls, plan engineering work, procure components and schedule corrective work at the next suitable maintenance cycle rather than defer the condition to routine monitoring.

14 Engineering Sketches and Reference Arrangement

The supplied wear-pattern illustrations distinguish a nominal, substantially symmetrical groove-contact pattern from localized sidewall wear that may occur under angular or off-centre rope tracking. The actual inspection photographs remain the primary evidence of the observed sheave condition. These diagrams are provided for explanatory and comparative context only and are not used as acceptance criteria in this report. Any allowable groove-wear limit, D:d ratio requirement (sheave diameter to rope diameter), or related acceptance criterion must be verified against the applicable OEM and/or client requirements before use.

14.1 Groove Wear Profile Context

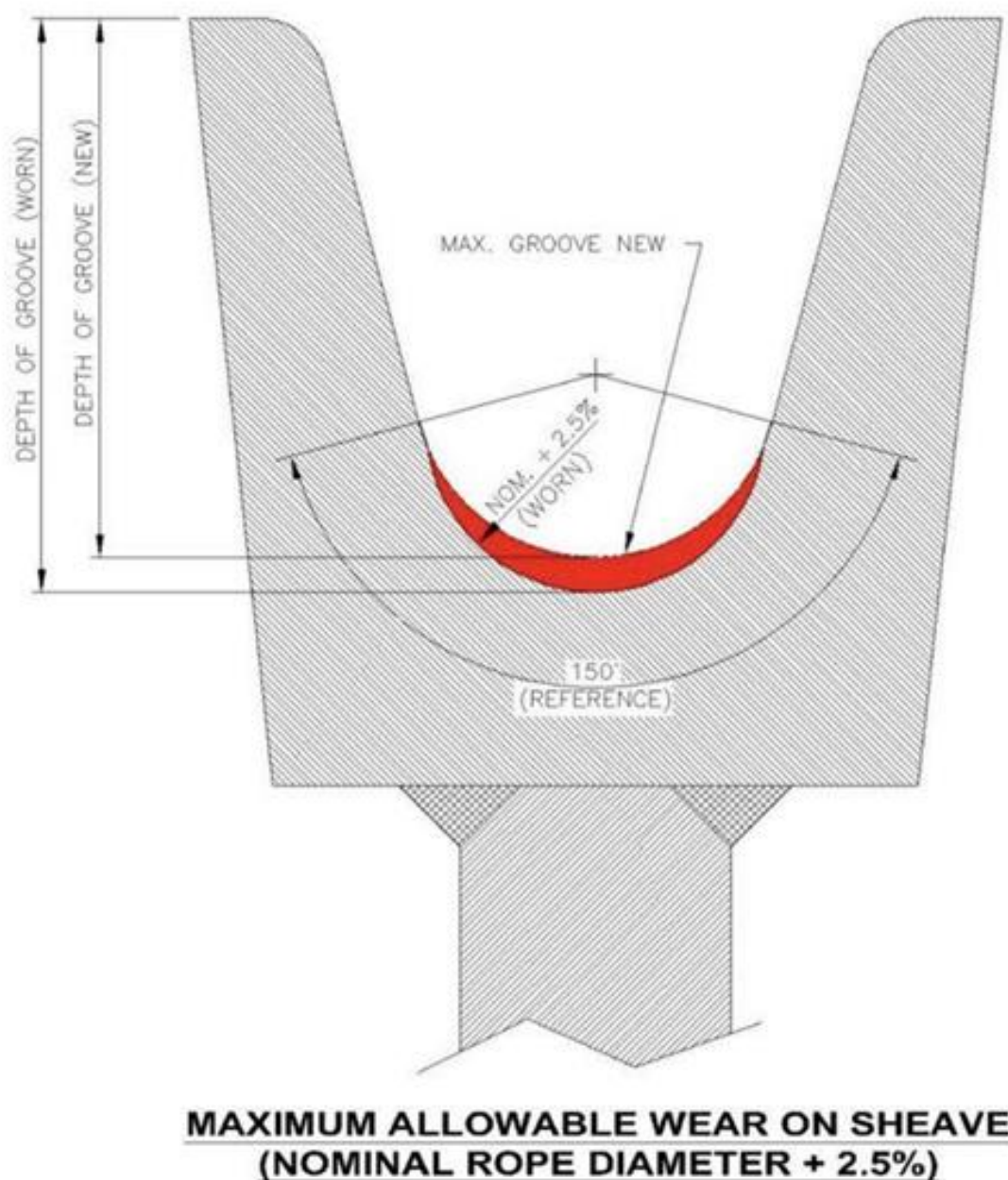


Figure 1 - CBASS-REF001 - Normal Wear Pattern

Nominal substantially symmetrical groove contact pattern.

LEFT-SIDE GROOVE WEAR

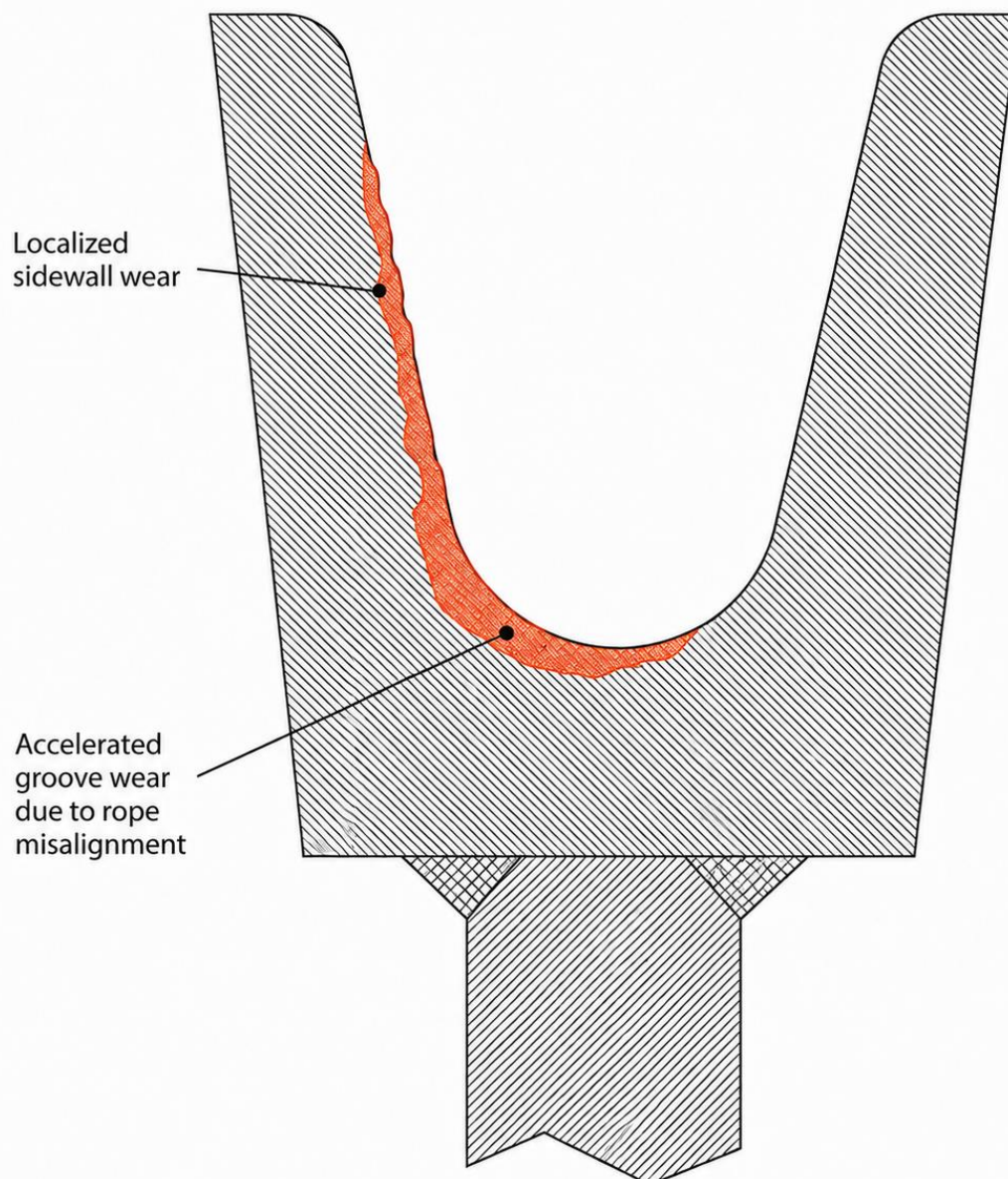


Figure 2 - CBASS-REF002 - Mis-aligned Groove Wear - Left Hand Side

Contextual example of localized left-side groove/sidewall wear under off-centre tracking.

RIGHT-SIDE GROOVE WEAR

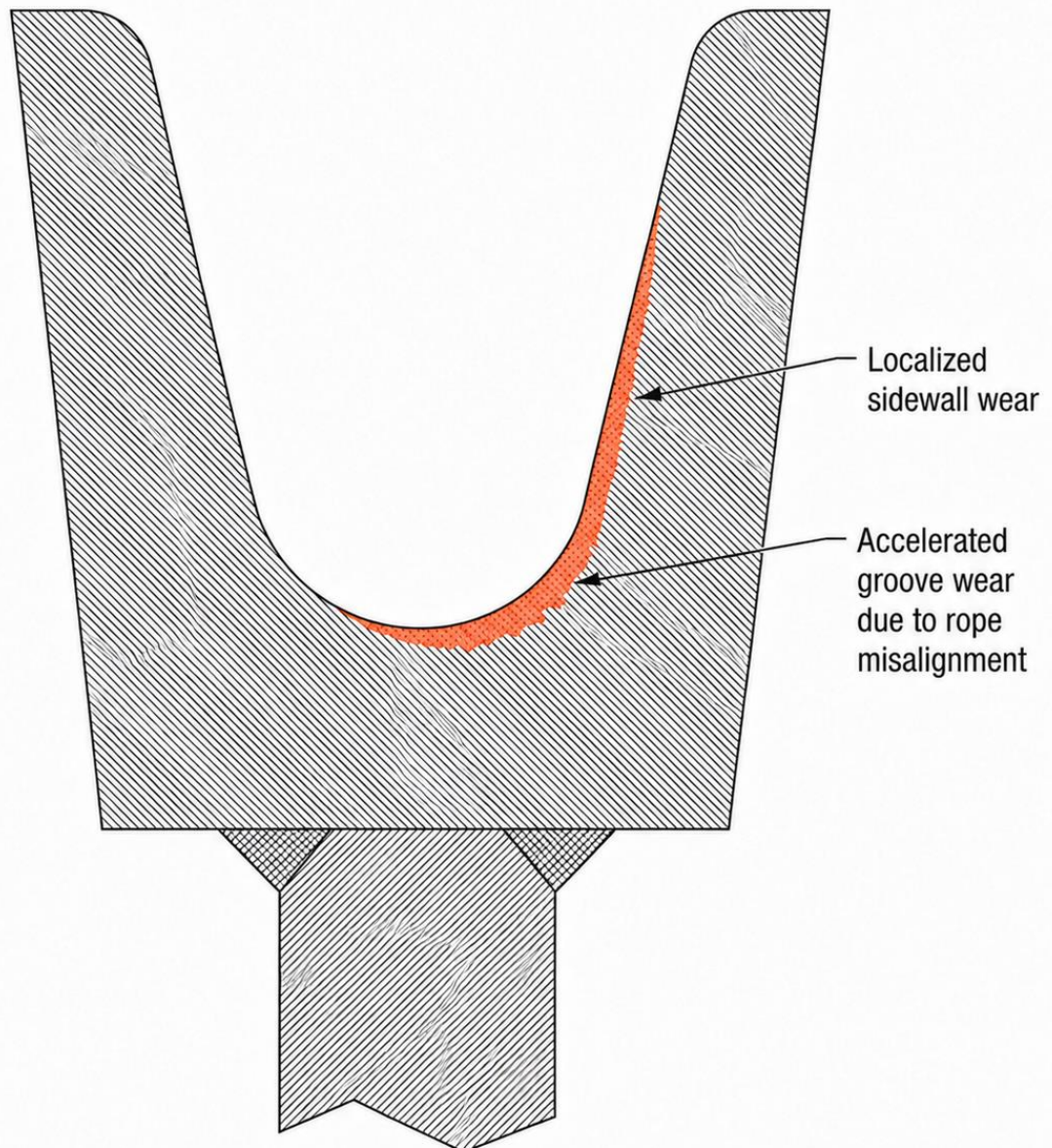


Figure 3 - CBASS-REF003 - Mis-aligned Groove Wear - Right Hand Side

Contextual example of localized right-side groove/sidewall wear under off-centre tracking.

14.2 Front-Rear Orientation Reference

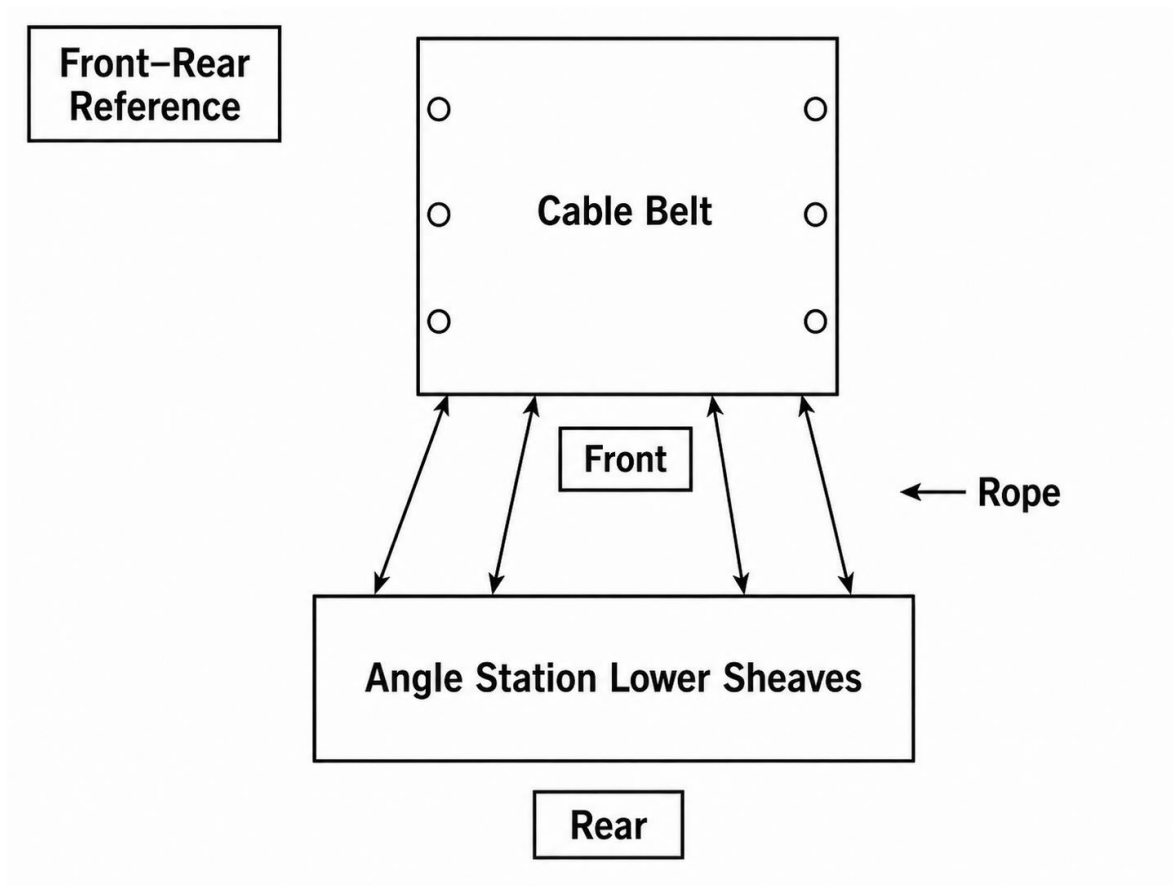


Figure 4 - CBASS-REF004 - Front-Rear Reference

Front is the side of the Lower Angle Station Sheaves facing the cable belt; rear is the opposite side.

14.3 Sheave Rope-Path Fleet-Angle Offset and Recommended Correction Concept

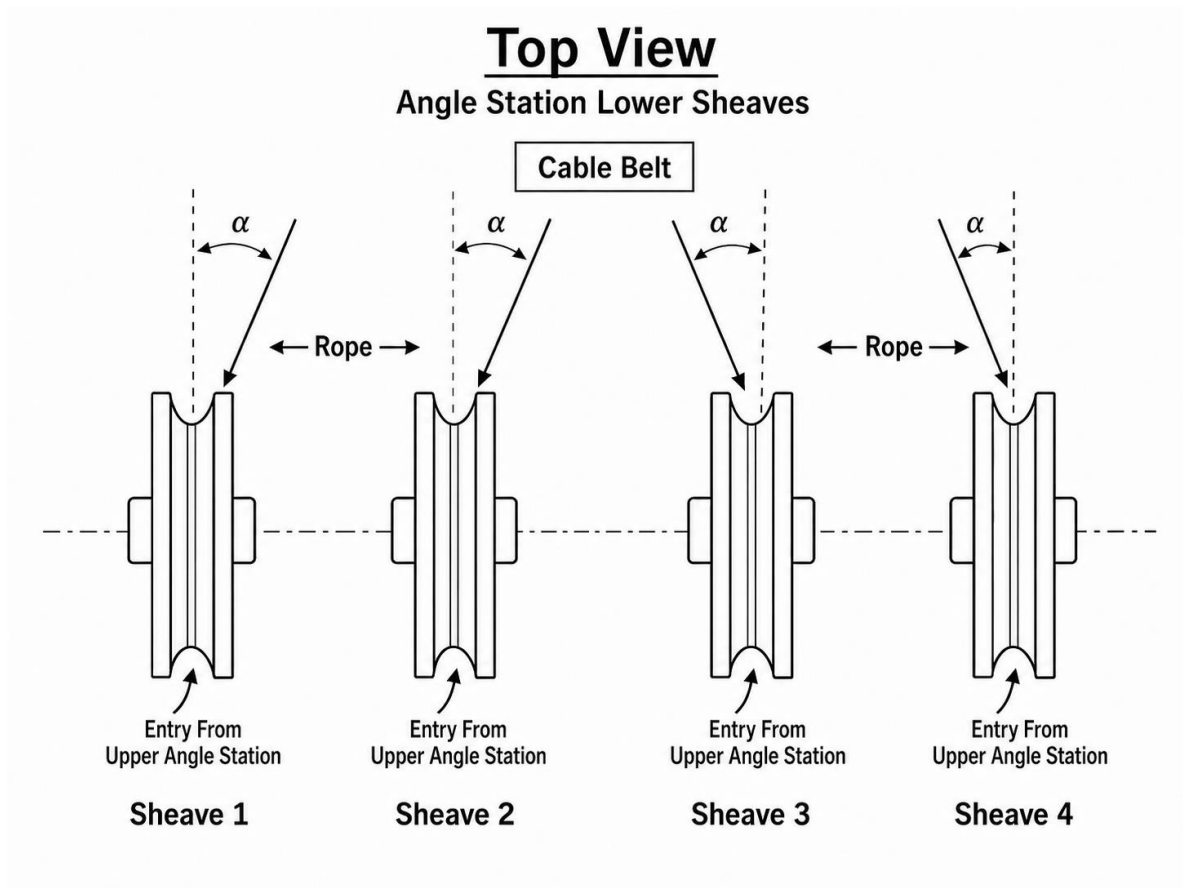


Figure 5 - CBASS-REF005 - Cable Belt Lower Sheaves Top View

Contextual top view showing non-zero rope approach/departure angle at the lower sheaves.

Proposed Configuration - Install Guide Sheaves

Rear View – Angle Station Lower Sheaves

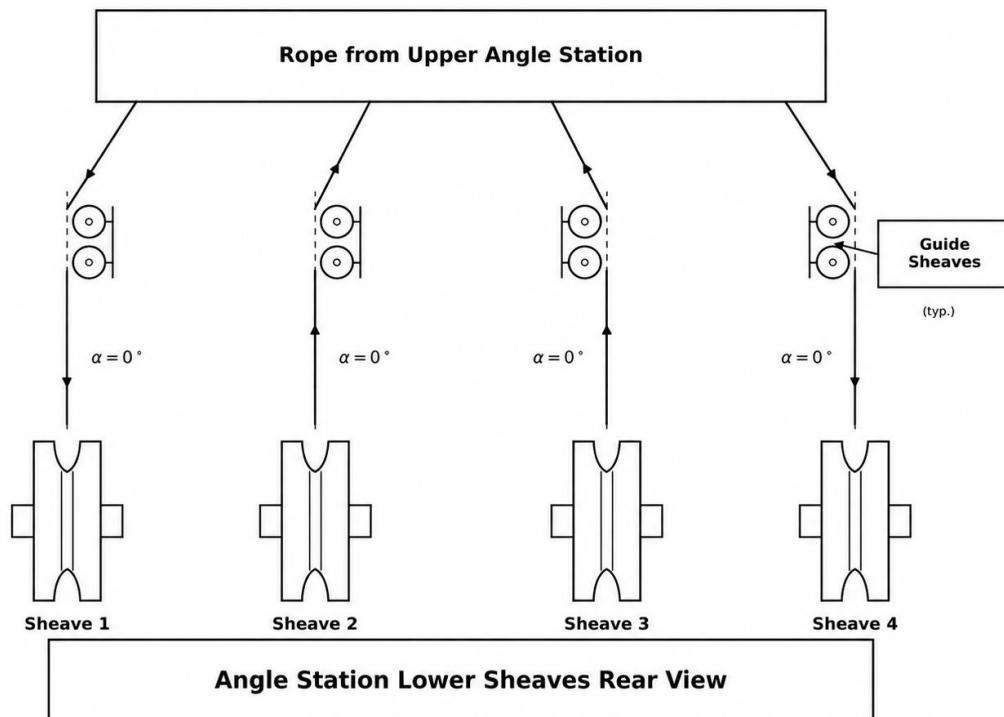


Figure 6 - CBASS-REF006 - Angle Station Lower Sheaves Rear View

Conceptual rear-view guide-sheave arrangement illustrating the functional requirement to straighten the rope path at the lower sheaves. Final guide-sheave quantity, position and configuration are subject to specialist detailed design.

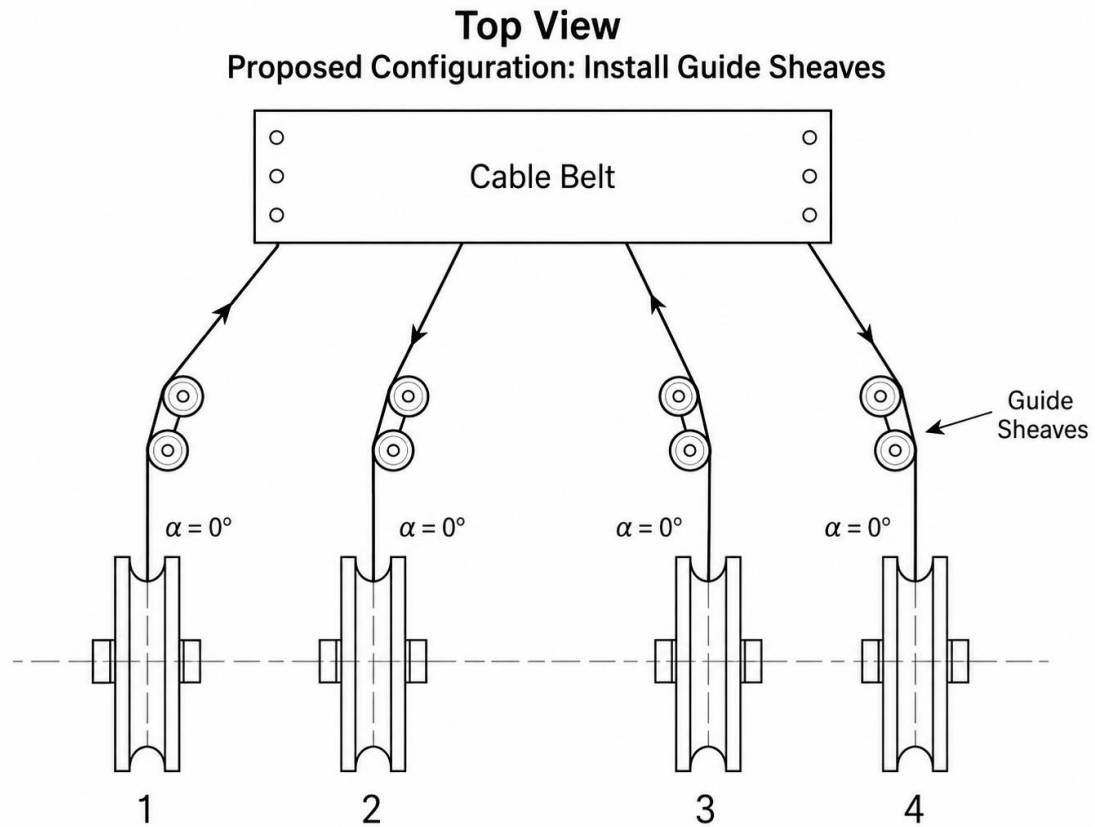


Figure 7 - CBASS-REF007 - Angle Station Lower Sheaves Top View

Conceptual top-view guide-sheave arrangement illustrating the intended approximately zero-degree final rope path into and out of the lower sheaves. The drawing provides context only and is not a construction or design guideline.

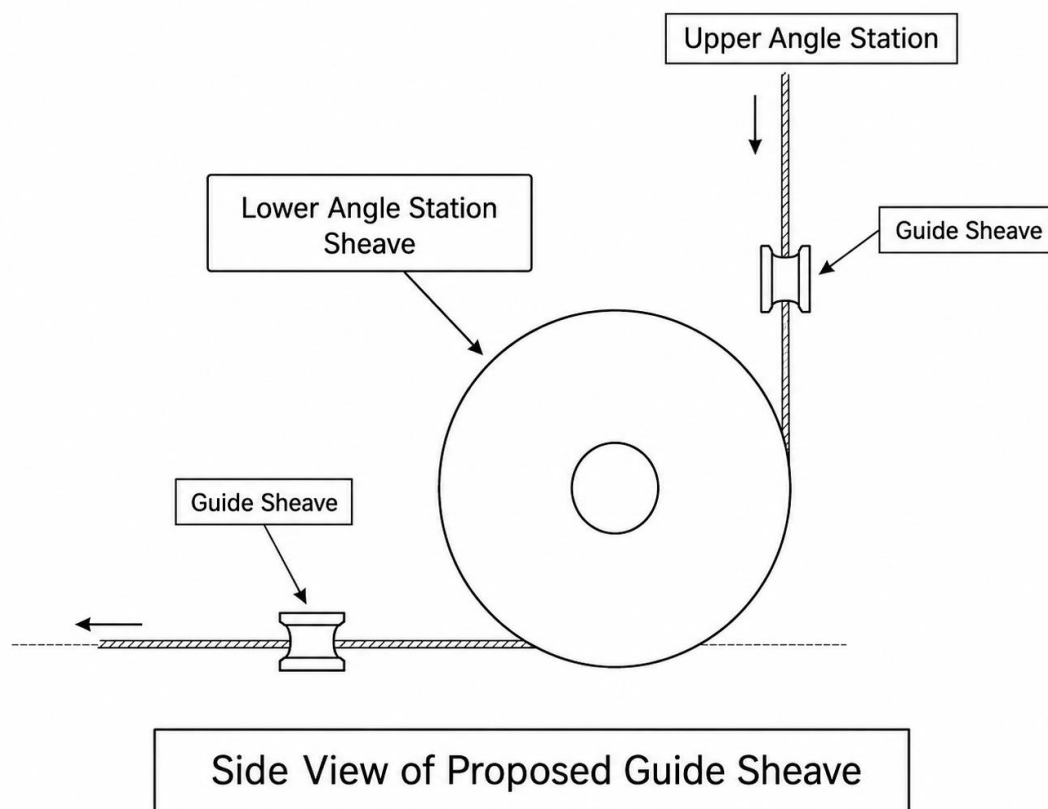


Figure 8 - CBASS-REF008 - Side View of Proposed Guide Sheaves

Conceptual side view of proposed guide control. The arrangement illustrates lateral rope guidance/deflection rather than a final wrap-around reeving layout; final geometry is subject to detailed design verification.

14.4 Angular Forces on Sheave Bearings

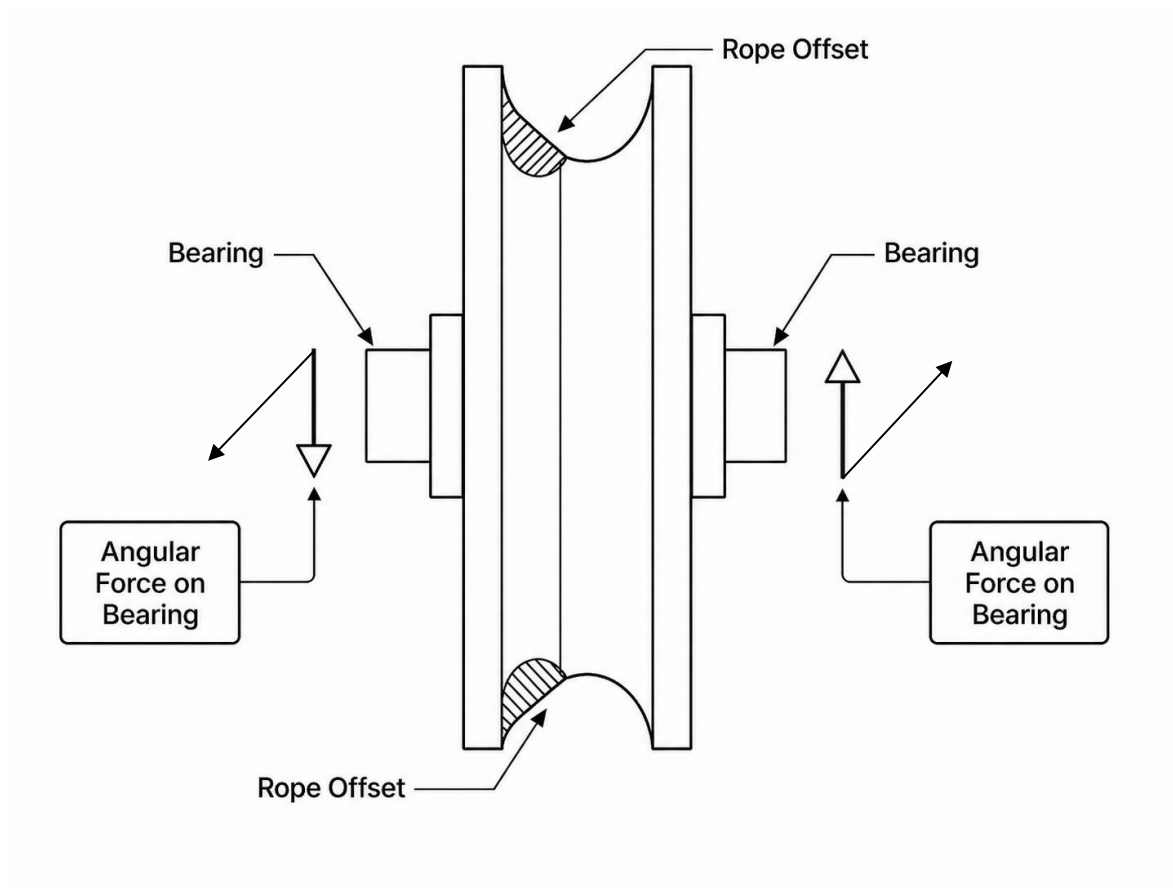


Figure 9 - CBASS-REF009 - Angular Forces on Sheave Bearings

Conceptual illustration of angular/lateral bearing loading associated with rope offset. Reference only.

14.5 Inspection Orientation Key

Reference	Description	Direction / datum	Source drawing / photograph
Front	Cable-belt side of the Lower Angle Station sheaves; photographic Front View datum	Looking from the Cable Belt/front side toward the rear	CBASS-REF004
Rear	Opposite side of the Lower Angle Station sheaves; photographic Rear-View datum	Looking from the rear toward the Cable Belt/front	CBASS-REF004
Sheave numbering	Lower Sheaves 1 to 4	Left-to-right as shown in report reference views	CBASS-REF005 to CBASS-REF007
Left/right perspective	Side-specific groove wear must always be read together with the stated Front View or Rear View.	A feature appearing left in Front View appears right in Rear View, and vice versa.	CBASS-REF004

15 Asset Descriptions

System arrangement: each Lower Angle Station sheave corresponds with an Upper Angle Station sheave. The angular rope transition between the two stations was physically observed. Each 45 mm rope follows an offset path relative to the main-sheave groove centre planes, producing non-zero entry and/or departure fleet angles. The same general asymmetric groove-wear condition was directly identified on all four Lower Angle Station sheaves. This common system geometry is assessed as the principal probable cause of the recurring lower-sheave wear and is used as the shared basis for the rope-tracking and engineering interpretation subsections below. Side-specific descriptions are viewpoint-dependent and are stated as Front View or Rear View where photographic evidence is used.

15.1 Cable Belt Angle Station Lower Sheave 1

1900 mm diameter sheave manufactured by Kimberley Engineering Works (Pty) Ltd. A top groove-depth reading of 104 mm was recorded. Asymmetric groove-side wear was directly observed during the inspection and recorded as a right-hand offset in the field notes; no usable groove photograph is available because of the camera technical issue.

Field	Entry
Asset	Cable Belt Angle Station Lower Sheave 1
Operational duty	Cable belt rope redirection at lower angle station
Location	Cable Belt Lower Angle Station
Principal components	Sheave, groove/rim, shaft/hub and bearing housings
Reference drawing / figure	CBASS-REF004 to CBASS-REF009
Inspection access state	Stopped / isolated state

15.2 Cable Belt Angle Station Lower Sheave 2

1900 mm diameter sheave manufactured by Kimberley Engineering Works (Pty) Ltd. A maximum top groove-depth reading of 106 mm was recorded. Front-view photographs CBASSL2-REF001 and CBASSL2-REF002 document asymmetric groove/sidewall contact wear. The wear appears on the left when viewed from the Front and on the right when viewed from the Rear; this is a viewing-perspective relationship, not a change in the physical wear location.

Field	Entry
Asset	Cable Belt Angle Station Lower Sheave 2
Operational duty	Cable belt rope redirection at lower angle station
Location	Cable Belt Lower Angle Station

Field	Entry
Principal components	Sheave, groove/rim, shaft/hub and bearing housings
Reference drawing / figure	CBASS-REF004 to CBASS-REF009
Inspection access state	Stopped / isolated state

15.3 Cable Belt Angle Station Lower Sheave 3

1900 mm diameter sheave manufactured by Kimberley Engineering Works (Pty) Ltd. A maximum top groove-depth reading of 105 mm was recorded. Front-view photographs CBASSL3-REF001 and CBASSL3-REF002 document the asymmetric groove/sidewall wear. The wear appears on the left when viewed from the Front and on the right when viewed from the Rear.

Field	Entry
Asset	Cable Belt Angle Station Lower Sheave 3
Operational duty	Cable belt rope redirection at lower angle station
Location	Cable Belt Lower Angle Station
Principal components	Sheave, groove/rim, shaft/hub and bearing housings
Reference drawing / figure	CBASS-REF004 to CBASS-REF009
Inspection access state	Stopped / isolated state

15.4 Cable Belt Angle Station Lower Sheave 4

Lower Angle Station sheave with OEM plate identifying Kimberley Engineering Works (Pty) Ltd., 1900 mm sheave diameter, 45 mm rope diameter and date of manufacture 01-2022. The separate plate records A = Rope Groove Depth 97 mm and B = Max Wear Depth 127 mm. A top groove-depth reading of 102 mm was recorded. Asymmetric groove-side wear was directly observed during the inspection and recorded in the field notes; no usable groove-wear photograph is available because of the camera technical issue.

Field	Entry
Asset	Cable Belt Angle Station Lower Sheave 4
Operational duty	Cable belt rope redirection at lower angle station
Location	Cable Belt Lower Angle Station
Principal components	Sheave, groove/rim, shaft/hub and bearing housings
Reference drawing / figure	CBASS-REF004 to CBASS-REF009
Inspection access state	Stopped / isolated state

16 Inspection Findings

16.1 Cable Belt Lower Angle Station Sheave 1

Defect ID	DEF-001	Asset	Cable Belt Lower Angle Station Sheave 1
Finding title	Asymmetric groove wear / rope-path offset	Location	Cable Belt Lower Angle Station

16.1.1 Asset Overview and General Condition

Cable Belt Lower Angle Station Sheave 1 | Cable Belt Lower Angle Station | Overall condition: Poor. Classification supplied: Level 2 - Moderate | P3 - Planned. Recorded groove-depth reading: 104 mm.

16.1.2 Inspection Observation

A top groove-depth reading of 104 mm was recorded. Asymmetric groove-side wear was directly observed during the inspection and recorded as a right-hand rope/groove offset in the field notes. A usable Sheave 1 groove photograph is not available because of the camera technical issue. The angular Upper-to-Lower rope transition was physically observed; the actual fleet-angle magnitude was not measured.

16.1.3 Rope Groove and Rim Condition

The recorded top groove depth is 104 mm. Relative to the supplied 97 mm nominal groove depth and 127 mm maximum wear depth, this represents approximately 7 mm of the 30 mm vertical wear range consumed, leaving approximately 23 mm before the maximum vertical wear-depth limit. The vertical depth is therefore within the supplied operating range. The separate concern is the directly observed asymmetric sidewall wear, which changes the rope support/contact condition and is not quantified by the single vertical depth reading.

16.1.4 Rope Tracking and Alignment

The 45 mm rope was observed transitioning through an angularly offset path between the corresponding Upper and Lower Angle Station positions. The final rope approach is not coincident with the lower sheave groove centre plane and therefore produces a non-zero entry fleet angle. For Sheave 1 the field notes record a right-hand rope/groove offset. The angular condition is physically established; the exact fleet angle and lateral offset remain to be quantified by survey.

16.1.5 Hub, Shaft and Bearing Housing

Internal bearing condition is not established by the visual groove inspection, and bearing arrangement, rope tension and condition-monitoring data were not supplied with this report package. The common load-path implications of off-centre rope contact are consolidated in Section 16.5. Available bearing vibration/temperature history and bearing arrangement data should be requested from the mine Condition Monitoring department where required.

16.1.6 Structural Components, Fasteners and Guards

The supplied photographs provide contextual views of surrounding supports only; no dedicated structural or NDT finding is assigned within this inspection scope.

16.1.7 Corrosion, Coating and Housekeeping

No material corrosion/coating finding is assigned from the supplied evidence package. Local surface contamination/operating residue is visible in contextual photographs.

16.1.8 Engineering Assessment

ENGINEERING INTERPRETATION

The directly observed asymmetric groove condition on Sheave 1 is consistent with the physically observed angular Upper-to-Lower rope path. The non-zero fleet-angle/off-centre tracking condition is assessed as the principal probable cause of the localized sidewall wear. Under rope tension, repeated contact against a groove flank can accelerate wear of both the rope and the sheave and can disturb the intended support of the rope strands. The current vertical groove depth remains within the supplied 127 mm maximum wear-depth limit, but the groove condition is nevertheless functionally Poor because the rope is no longer being supported optimally through the groove centreline. The common bearing-load mechanism and system-level recurrence risk are addressed in Section 16.5.

16.1.9 Defect Classification and Recommended Action

Defect ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-001	Poor	Level 2 - Moderate	P3 - Planned	Continue only under the P3 planned-maintenance framework and the mine's operational/condition-monitoring controls. Complete the full Angle Station geometry survey and implement the recommended engineered guide control at both entry and departure of the relevant main sheaves. When the corresponding rope-path correction is commissioned, replace the existing worn Lower Angle Station main sheave with a new sheave as part of the same intervention; do not commission the corrected rope path through the existing distorted Lower groove. If bearing or other condition indicators deteriorate beyond mine-defined limits before the planned intervention, replace the affected sheave at the earliest practicable maintenance opportunity. See Section 20.	Mechanical / Engineering	Mine condition-monitoring review; as-found/as-left survey readings; approved guide-sheave/support design; new main-sheave installation record; post-installation centred tracking verification; repeat groove/rope wear records.

16.1.10 Photographic Evidence

No usable Sheave 1 groove photograph is available because of the camera technical issue encountered during the inspection. The asymmetric groove-side wear was physically observed and recorded in the field notes.

16.2 Cable Belt Lower Angle Station Sheave 2

Defect ID	DEF-002	Asset	Cable Belt Lower Angle Station Sheave 2
Finding title	Asymmetric groove wear / rope-path offset	Location	Cable Belt Lower Angle Station

16.2.1 Asset Overview and General Condition

Cable Belt Lower Angle Station Sheave 2 | Cable Belt Lower Angle Station | Overall condition: Poor. Classification supplied: Level 2 - Moderate | P3 - Planned. Recorded groove-depth reading: 106 mm.

16.2.2 Inspection Observation

A maximum top groove-depth reading of 106 mm was recorded. Front-view photographs CBASSL2-REF001 and CBASSL2-REF002 show a concentrated polished/contact band and localized asymmetric sidewall/groove-base wear. In these Front View photographs, the worn groove side appears on the left; the same physical wear location would appear on the right when viewed from the Rear. This reflects viewing perspective only; the physical wear location is unchanged. The angular rope transition was physically observed, but the actual entry fleet angle was not measured.

16.2.3 Rope Groove and Rim Condition

The recorded top groove depth is 106 mm. Relative to the supplied 97 mm nominal groove depth and 127 mm maximum wear depth, this represents approximately 9 mm of the 30 mm vertical wear range consumed, leaving approximately 21 mm before the maximum vertical wear-depth limit. The vertical depth remains within the supplied operating range. Front-view photographs show localized asymmetric sidewall/groove-base contact, which is a separate lateral wear condition and is not represented by the single depth value.

16.2.4 Rope Tracking and Alignment

The corresponding Upper-to-Lower rope path was physically observed to be angularly offset, so the rope does not approach Sheave 2 on the groove centre plane. This produces a non-zero entry fleet angle and preferential side contact. CBASSL2-REF001 and CBASSL2-REF002 are Front View photographs: the wear appears on the left in that viewpoint and corresponds to the right-hand side when the same physical groove is viewed from the Rear. The exact fleet angle and lateral offset were not measured.

16.2.5 Hub, Shaft and Bearing Housing

Internal bearing condition is not established by the visual groove inspection, and bearing arrangement, rope tension and condition-monitoring data were not supplied with this report package. The common load-path implications of off-centre rope contact are consolidated in Section 16.5. Available bearing vibration/temperature history and bearing arrangement data should be requested from the mine Condition Monitoring department where required.

16.2.6 Structural Components, Fasteners and Guards

The supplied photographs provide contextual views of surrounding supports only; no dedicated structural or NDT finding is assigned within this inspection scope.

16.2.7 Corrosion, Coating and Housekeeping

No material corrosion/coating finding is assigned from the supplied evidence package. Local surface contamination/operating residue is visible in contextual photographs.

16.2.8 Engineering Assessment

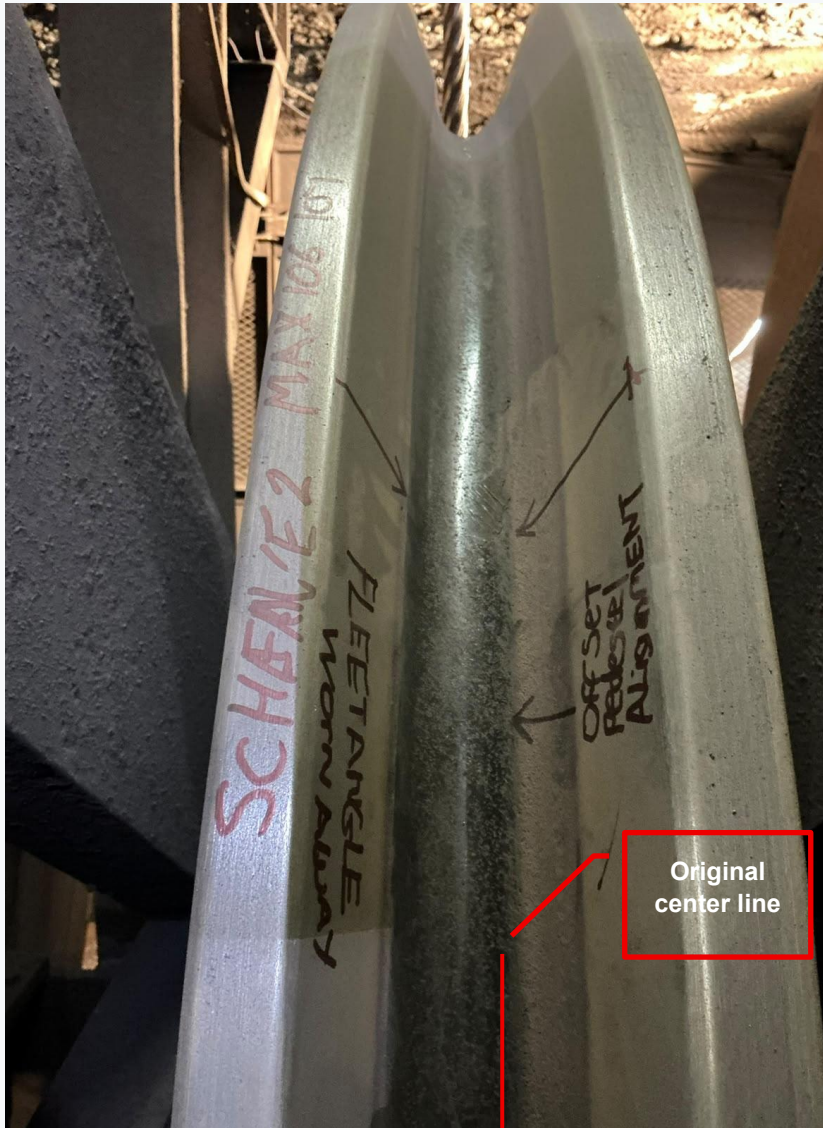
ENGINEERING INTERPRETATION

The front-view photographs provide direct evidence of localized asymmetric groove contact on Sheave 2. In combination with the physically observed angular Upper-to-Lower rope path, the non-zero fleet-angle/off-centre tracking condition is assessed as the principal probable cause of the wear pattern. The rope is not supported continuously through the intended groove centreline and instead bears preferentially against the groove flank, increasing localized contact stress and the potential for accelerated rope and sheave wear. The 106 mm vertical depth remains inside the supplied maximum wear-depth limit, but the asymmetric profile justifies the Poor / Level 2 / P3 classification described in Section 13. The common bearing-load effect and system recurrence mechanism are consolidated in Section 16.5.

16.2.9 Defect Classification and Recommended Action

Defect ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-002	Poor	Level 2 - Moderate	P3 - Planned	Continue only under the P3 planned-maintenance framework and the mine's operational/condition-monitoring controls. Complete the full Angle Station geometry survey and implement the recommended engineered guide control at both entry and departure of the relevant main sheaves. When the corresponding rope-path correction is commissioned, replace the existing worn Lower Angle Station main sheave with a new sheave as part of the same intervention; do not commission the corrected rope path through the existing distorted Lower groove. If bearing or other condition indicators deteriorate beyond mine-defined limits before the planned intervention, replace the affected sheave at the earliest practicable maintenance opportunity. See Section 20.	Mechanical / Engineering	Mine condition-monitoring review; as-found/as-left survey readings; approved guide-sheave/support design; new main-sheave installation record; post-installation centred tracking verification; repeat groove/rope wear records.

16.2.10 Photographic Evidence - CBASSL2-REF001

Photo ID
CBASSL2-REF001Defect ID
DEF-002Asset
Cable Belt Lower Angle Station Sheave 2Component / location
Rope groove / lower angle station sheaveView direction
Front viewDate / time
Not supplied

Caption
Front-view contextual groove view showing the asymmetric polished/contact band and localized groove-side wear on Lower Angle Station Sheave 2.

Photographic evidence sheet - CBASSL2-REF001 / DEF-002

Observation	Engineering assessment	Recommended action
Front-view contextual groove view showing the asymmetric polished/contact band and localized groove-side wear on Lower Angle Station Sheave 2.	The photograph directly supports an asymmetric groove-contact condition. Because this is a Front View image, the wear appears on the left-hand side; the same physical groove location appears on the right-hand side from the Rear. This reflects viewing perspective only; the physical wear location is unchanged. In the context of the physically observed angular rope path, the non-zero fleet-angle/off-centre tracking condition is assessed as the principal probable cause. The photograph does not quantify the fleet angle or bearing load.	P3 planned: complete the full rope-path geometry survey and specialist detailed design; implement entry/departure guide control and replace the existing worn Lower main sheave when the correction is commissioned; verify centred rope tracking and trend groove, rope and bearing condition.

16.2.11 Photographic Evidence - CBASSL2-REF002



Photo ID
CBASSL2-REF002

Defect ID
DEF-002

Asset
Cable Belt Lower Angle Station Sheave 2

Component / location
Rope groove / lower angle station sheave

View direction
Front view

Date / time
Not supplied

Caption
Front-view close/contextual groove view showing asymmetric polished/contact wear concentrated along the groove sidewall and groove base on Lower Angle Station Sheave 2.

Photographic evidence sheet - CBASSL2-REF002 / DEF-002

Observation	Engineering assessment	Recommended action
Front-view close/contextual groove view showing asymmetric polished/contact wear concentrated along the groove sidewall and groove base on Lower Angle Station Sheave 2.	The photograph directly supports an asymmetric groove-contact condition. Because this is a Front View image, the wear appears on the left-hand side; the same physical groove location appears on the right-hand side from the Rear. This reflects viewing perspective only; the physical wear location is unchanged. In the context of the physically observed angular rope path, the non-zero fleet-angle/off-centre tracking condition is assessed as the principal probable cause. The photograph does not quantify the fleet angle or bearing load.	P3 planned: complete the full rope-path geometry survey and specialist detailed design; implement entry/departure guide control and replace the existing worn Lower main sheave when the correction is commissioned; verify centred rope tracking and trend groove, rope and bearing condition.

16.3 Cable Belt Lower Angle Station Sheave 3

Defect ID	DEF-003	Asset	Cable Belt Lower Angle Station Sheave 3
Finding title	Asymmetric groove wear / rope-path offset	Location	Cable Belt Lower Angle Station

16.3.1 Asset Overview and General Condition

Cable Belt Lower Angle Station Sheave 3 | Cable Belt Lower Angle Station | Overall condition: Poor. Classification supplied: Level 2 - Moderate | P3 - Planned. Recorded groove-depth reading: 105 mm.

16.3.2 Inspection Observation

A maximum top groove-depth reading of 105 mm was recorded. The Front View photographs CBASSL3-REF001 and CBASSL3-REF002 show the Sheave 3 overview and a close-up of the asymmetric groove-wear/contact pattern. In Front View the worn groove side appears on the left; the same physical wear location appears on the right when viewed from the Rear. The angular rope transition was physically observed, but the actual entry fleet angle was not measured. The bearing temperature-monitoring sensor for Sheave 3 was observed not attached to the sheave bearing and requires reattachment and functional verification.

16.3.3 Rope Groove and Rim Condition

The recorded top groove depth is 105 mm. Relative to the supplied 97 mm nominal groove depth and 127 mm maximum wear depth, this represents approximately 8 mm of the 30 mm vertical wear range consumed, leaving approximately 22 mm before the maximum vertical wear-depth limit. The vertical depth remains within the supplied operating range. The Front View photographs show localized asymmetric sidewall/groove-base contact that is not represented by the single depth reading.

16.3.4 Rope Tracking and Alignment

The corresponding Upper-to-Lower rope path was physically observed to be angularly offset, so the rope approaches Sheave 3 at a non-zero entry fleet angle instead of remaining centred on the groove plane. CBASSL3-REF001 and CBASSL3-REF002 are Front View photographs: the wear appears on the left from that viewpoint and corresponds to the right-hand side when viewed from the Rear. The exact fleet angle and lateral offset were not measured.

16.3.5 Hub, Shaft and Bearing Housing

Internal bearing condition is not established by the visual groove inspection, and bearing arrangement and rope tension were not supplied with this report package. The Sheave 3 bearing temperature-monitoring sensor was observed not attached to the bearing; bearing-temperature trending should therefore not be relied upon until the sensor is correctly reattached and functionally verified. The common load-path implications of off-centre rope contact are consolidated in Section 16.5. Available vibration history and other bearing condition-monitoring information should be reviewed through the mine Condition Monitoring department.

16.3.6 Structural Components, Fasteners and Guards

The supplied photographs provide contextual views of surrounding supports only; no dedicated structural or NDT finding is assigned within this inspection scope.

16.3.7 Corrosion, Coating and Housekeeping

No material corrosion/coating finding is assigned from the supplied evidence package. Local surface contamination/operating residue is visible in contextual photographs.

16.3.8 Engineering Assessment

ENGINEERING INTERPRETATION

The Front View photographs provide direct evidence of the asymmetric groove-contact condition on Sheave 3. In combination with the physically observed angular Upper-to-Lower rope path, the non-zero fleet-angle/off-centre tracking condition is assessed as the principal probable cause of the wear pattern. The localized groove-flank contact can accelerate wear of the 45 mm rope and the sheave while disturbing the intended centred support of the rope. The 105 mm vertical groove depth remains within the supplied maximum wear-depth limit, but the asymmetric profile justifies the Poor / Level 2 / P3 classification described in Section 13. The detached temperature-monitoring sensor is a separate condition monitoring deficiency and should be rectified so that bearing temperature can be reliably trended. The common bearing-load mechanism is consolidated in Section 16.5.

16.3.9 Defect Classification and Recommended Action

Defect ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-003	Poor	Level 2 - Moderate	P3 - Planned	Continue only under the P3 planned-maintenance framework and the mine's operational/condition-monitoring controls. Complete the full Angle Station geometry survey and implement the recommended engineered guide control at both entry and departure of the relevant main sheaves. When the corresponding rope-path correction is commissioned, replace the existing worn Lower Angle Station main sheave with a new sheave as part of the same intervention; do not commission the corrected rope path through the existing distorted Lower groove. If bearing or other condition indicators deteriorate beyond mine-defined limits before the planned intervention, replace the affected sheave at the earliest practicable maintenance opportunity. Reattach and functionally verify the Sheave 3 bearing temperature-monitoring sensor before relying on temperature trending. See Section 20.	Mechanical / Engineering	Temperature-sensor reattachment/function check; mine condition-monitoring review; as-found/as-left survey readings; approved guide-sheave/support design; new main-sheave installation record; post-installation centred tracking verification; repeat groove/rope wear records.

16.3.10 Photographic Evidence - CBASSL3-REF001

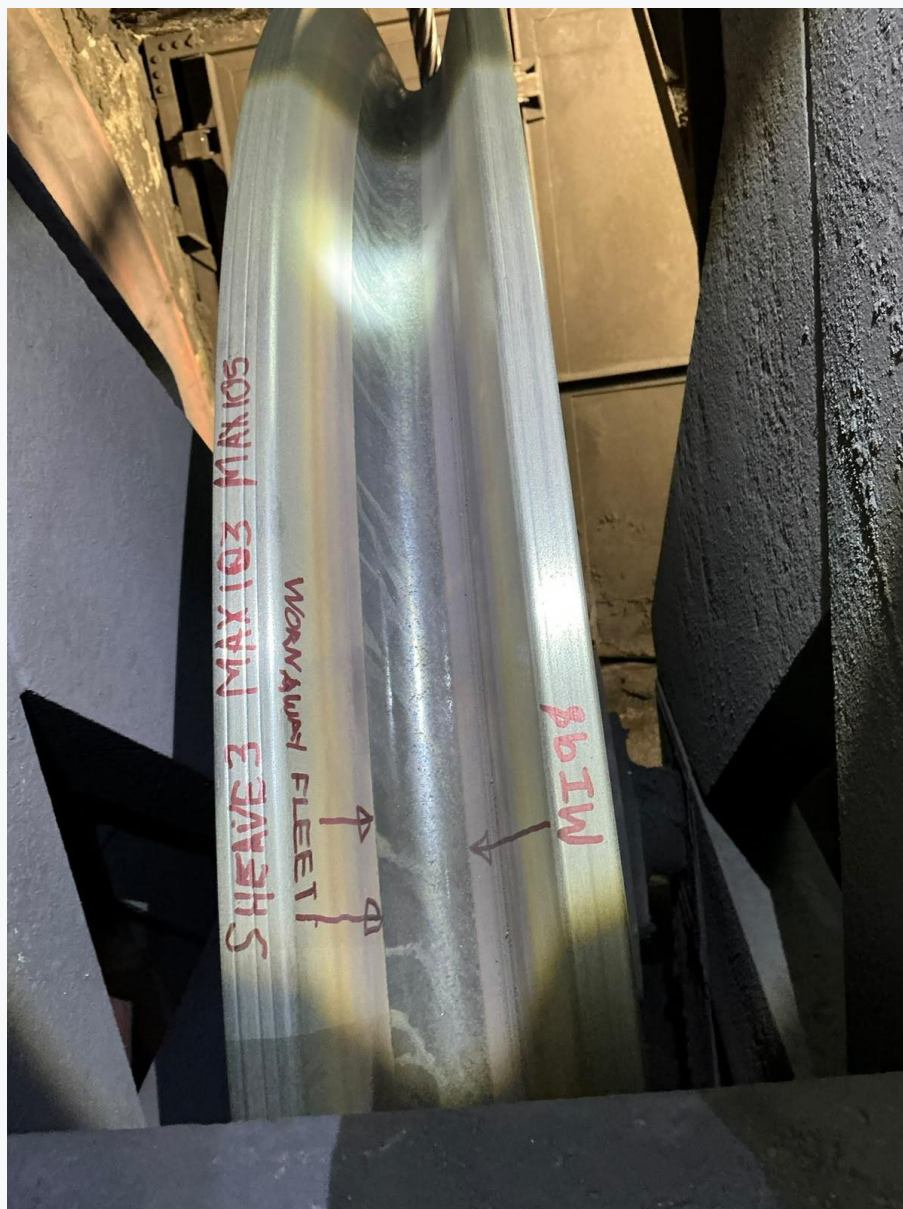


Photo ID
CBASSL3-REF001

Defect ID
DEF-003

Asset
Cable Belt Lower Angle Station
Sheave 3

Component / location
Rope groove / lower angle station
sheave

View direction
Front view

Date / time
Not supplied

Caption
Front-view overview of Lower Angle
Station Sheave 3 showing the
asymmetric groove-wear/contact
pattern and the marked inspection
context.

Photographic evidence sheet - CBASSL3-REF001 / DEF-003

Observation	Engineering assessment	Recommended action
Front-view overview of Lower Angle Station Sheave 3 showing the asymmetric groove-wear/contact pattern and the marked inspection context.	The Front View photograph directly supports the asymmetric groove-contact condition on Sheave 3. The wear appears on the left-hand side in this viewpoint and is the same physical location that appears on the right-hand side from the Rear. In combination with the physically observed angular rope path, the non-zero fleet-angle/off-centre tracking condition is assessed as the principal probable cause. The photograph does not quantify the fleet angle or bearing load.	P3 planned: complete the full rope-path geometry survey and specialist detailed design; implement entry/departure guide control and replace the existing worn Lower main sheave when the correction is commissioned; verify centred rope tracking and trend groove, rope and bearing condition.

16.3.11 Photographic Evidence - CBASSL3-REF002



Photo ID
CBASSL3-REF002

Defect ID
DEF-003

Asset
Cable Belt Lower Angle Station
Sheave 3

Component / location
Rope groove / lower angle station
sheave

View direction
Front view

Date / time
Not supplied

Caption
Front-view close-up of CBASSL3-REF001 showing the localized asymmetric groove-wear/contact pattern on Lower Angle Station Sheave 3.

Photographic evidence sheet - CBASSL3-REF002 / DEF-003

Observation	Engineering assessment	Recommended action
Front-view close-up of CBASSL3-REF001 showing the localized asymmetric groove-wear/contact pattern on Lower Angle Station Sheave 3.	The Front View photograph directly supports the asymmetric groove-contact condition on Sheave 3. The wear appears on the left-hand side in this viewpoint and is the same physical location that appears on the right-hand side from the Rear. In combination with the physically observed angular rope path, the non-zero fleet-angle/off-centre tracking condition is assessed as the principal probable cause. The photograph does not quantify the fleet angle or bearing load.	P3 planned: complete the full rope-path geometry survey and specialist detailed design; implement entry/departure guide control and replace the existing worn Lower main sheave when the correction is commissioned; verify centred rope tracking and trend groove, rope and bearing condition.

16.4 Cable Belt Lower Angle Station Sheave 4

Defect ID	DEF-004	Asset	Cable Belt Lower Angle Station Sheave 4
Finding title	Asymmetric groove wear / rope-path offset	Location	Cable Belt Lower Angle Station

16.4.1 Asset Overview and General Condition

Cable Belt Lower Angle Station Sheave 4 | Cable Belt Lower Angle Station | Overall condition: Poor. Classification supplied: Level 2 - Moderate | P3 - Planned. Recorded groove-depth reading: 102 mm.

16.4.2 Inspection Observation

A top groove-depth reading of 102 mm was recorded. Asymmetric groove-side wear was directly observed during the inspection and recorded in the field notes. CBASSL4-REF001 confirms the sheave manufacturer, 1900 mm sheave diameter, 45 mm rope diameter and date of manufacture 01-2022. CBASSL4-REF002 records A = Rope Groove Depth 97 mm and B = Max Wear Depth 127 mm. A usable Sheave 4 groove-wear photograph is not available because of the camera technical issue. The angular rope transition was physically observed; the actual entry fleet angle was not measured. The bearing temperature-monitoring sensor for Sheave 4 was observed not attached to the sheave bearing and requires reattachment and functional verification.

16.4.3 Rope Groove and Rim Condition

OEM/asset plate evidence records A = 97 mm nominal rope-groove depth and B = 127 mm maximum wear depth. The recorded top groove depth is 102 mm, representing approximately 5 mm of the 30 mm vertical wear range consumed and approximately 25 mm remaining before the maximum vertical wear-depth limit. The vertical depth therefore remains within the supplied operating range. Asymmetric groove-side wear was nevertheless directly observed during inspection; the absence of a usable groove photograph is a photographic-record limitation caused by the camera technical issue, not an indication that the wear was not observed.

16.4.4 Rope Tracking and Alignment

The corresponding Upper-to-Lower rope transition was physically observed to be angularly offset and therefore produces a non-zero entry fleet angle at Sheave 4. The field notes record the side-specific offset observed during inspection. Because no usable groove photograph is available, the side designation is retained as a field observation rather than a photographic comparison. The angular condition is established qualitatively; the exact fleet angle and lateral offset were not measured.

16.4.5 Hub, Shaft and Bearing Housing

Internal bearing condition is not established by the visual groove inspection, and bearing arrangement and rope tension were not supplied with this report package. The Sheave 4 bearing temperature-monitoring sensor was observed not attached to the bearing; bearing-temperature trending should therefore not be relied upon until the sensor is correctly reattached and functionally verified. The common load-path implications of off-centre rope contact are consolidated in Section 16.5. Available vibration history and other bearing condition-monitoring information should be reviewed through the mine Condition Monitoring department.

16.4.6 Structural Components, Fasteners and Guards

The supplied photographs provide contextual views of surrounding supports only; no dedicated structural or NDT finding is assigned within this inspection scope.

16.4.7 Corrosion, Coating and Housekeeping

No material corrosion/coating finding is assigned from the supplied evidence package. Local surface contamination/operating residue is visible in contextual photographs.

16.4.8 Engineering Assessment

ENGINEERING INTERPRETATION

The directly observed asymmetric groove condition on Sheave 4 is consistent with the same system geometry affecting the other Lower Angle Station sheaves. The non-zero fleet-angle/off-centre tracking condition is assessed as the principal probable cause of the recurring sidewall wear. The 102 mm groove depth remains within the supplied 127 mm maximum wear-depth limit, but vertical depth alone does not capture the lateral groove profile or its effect on rope support. The separate Upper Angle Station Sheave No. 4 failure evidence in Appendix F is relevant system context; the bearing collapse, displacement, debris and post-extraction

scoring/smearing establish the failure mode and damage pattern, while a dedicated root-cause analysis would still be required to prove sole causation for that individual failure. The detached Lower Sheave 4 temperature-monitoring sensor is a separate condition-monitoring deficiency and should be rectified. The common bearing-load and recurrence mechanism is consolidated in Section 16.5.

16.4.9 Defect Classification and Recommended Action

Defect ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-004	Poor	Level 2 - Moderate	P3 - Planned	Continue only under the P3 planned-maintenance framework and the mine's operational/condition-monitoring controls. Complete the full Angle Station geometry survey and implement the recommended engineered guide control at both entry and departure of the relevant main sheaves. When the corresponding rope-path correction is commissioned, replace the existing worn Lower Angle Station main sheave with a new sheave as part of the same intervention; do not commission the corrected rope path through the existing distorted Lower groove. If bearing or other condition indicators deteriorate beyond mine-defined limits before the planned intervention, replace the affected sheave at the earliest practicable maintenance opportunity. Reattach and functionally verify the Sheave 4 bearing temperature-monitoring sensor before relying on temperature trending. See Section 20.	Mechanical / Engineering	Temperature-sensor reattachment/function check; mine condition-monitoring review; as-found/as-left survey readings; approved guide-sheave/support design; new main-sheave installation record; post-installation centred tracking verification; repeat groove/rope wear records.

16.4.10 Photographic Evidence

No usable Lower Angle Station Sheave 4 groove-wear photograph is available because of the camera technical issue encountered during the inspection. The asymmetric groove-side wear was physically observed and recorded; asset and groove-limit specification photographs are included in Appendix A.

16.5 Common System Engineering Observation - Rope Path, Groove Wear and Bearing Load

All four Lower Angle Station sheaves were physically inspected and the same general asymmetric groove-side wear condition was identified. The angular transition of the 45 mm cable-belt ropes between the Upper and Lower Angle Stations was also physically observed and is supported by the system layout. On the balance of this direct geometry observation and the recurring lower-sheave wear pattern, the non-zero fleet-angle/off-centre rope-tracking condition is assessed as the principal probable cause of the asymmetric Lower Angle Station wear. The outstanding quantitative survey is required to measure the magnitude of the condition and to define the final corrective geometry; it is not the sole basis for establishing that the angular condition exists.

The primary functional consequence is that the rope does not remain continuously centred and supported in the intended groove plane. Instead, the tensioned rope bears preferentially against a groove flank, creating localized rubbing/abrasive contact. As the groove profile deteriorates, the rope can be subjected to localized compression or pinching of strands and increased contact stress, contributing to fatigue and accelerated rope wear even while the vertical groove depth remains inside the supplied 127 mm maximum wear-depth limit.

The same off-centre line of action changes the sheave load path. In simplified terms: rope force applied away from the intended sheave centre plane creates a lateral reaction and an overturning/tilting moment, which must be reacted by the shaft and bearing pair and can result in unequal bearing reactions. The magnitude has not been calculated because rope tension, bearing type/spacing and quantitative alignment data were not included. Figure 9 (CBASS-REF009) is therefore a conceptual explanation of the load path, not a bearing-load calculation. During the Lower Angle Station inspection, the bearing temperature-monitoring sensors on Sheaves 3 and 4 were found not attached

to their bearings. These sensors should be correctly reattached and functionally verified before bearing-temperature data is relied upon. Available vibration and other bearing condition-monitoring data should be reviewed through the mine Condition Monitoring department.

Upper Angle Station Sheave No. 4 provides relevant corroborating system evidence. The supplied photographs directly establish bearing collapse/displacement, metallic debris and severe post-failure scoring/smearing in the bearing/hub region, together with asymmetric groove/contact features and an in-situ angular rope/sheave arrangement. These observations support the relevance of abnormal off-centre loading as part of the wider system mechanism, but a dedicated failure/root-cause analysis would still be required to assign sole causal responsibility for that individual bearing collapse. Detailed inspection of Upper Sheaves 1-3 was not completed because the plant was returned to production; their detailed condition remains to be verified, although the angular rope paths between the Upper and Lower stations were physically observed.

17 Summary of Defects

Table 18 - Consolidated defect register

Defect ID	Asset	Component / location	Defect / observation	Condition	Severity	Priority	Evidence	Responsible discipline	Status
DEF-001	Cable Belt Lower Angle Station Sheave 1	Rope groove / tracking	Directly observed asymmetric groove wear / rope-path offset; groove depth 104 mm	Poor	Level 2	P3	Direct field observation / ISS inspection records / MEAS-001	Mechanical / Engineering	Open
DEF-002	Cable Belt Lower Angle Station Sheave 2	Rope groove / tracking	Asymmetric groove wear / rope-path offset; groove depth 106 mm; Front View wear appears left (Rear View right)	Poor	Level 2	P3	CBASSL2-REF001; CBASSL2-REF002	Mechanical / Engineering	Open
DEF-003	Cable Belt Lower Angle Station Sheave 3	Rope groove / tracking	Asymmetric groove wear / rope-path offset; groove depth 105 mm; Front View wear appears left (Rear View right)	Poor	Level 2	P3	CBASSL3-REF001; CBASSL3-REF002	Mechanical / Engineering	Open
DEF-004	Cable Belt Lower Angle Station Sheave 4	Rope groove / tracking	Directly observed asymmetric groove wear / rope-path offset; groove depth 102 mm	Poor	Level 2	P3	Direct field observation / CBASSL4-REF001 / CBASSL4-REF002 / MEAS-004	Mechanical / Engineering	Open

18 Inspection Observation Log

Table 19 - Inspection observation and evidence log

Observation ID	Asset	Location	Observation	Evidence reference	Condition rating	Related defect ID	Status
OBS-001	Cable Belt Lower Angle Station Sheave 1	Rope groove / rope path	Top groove depth 104 mm. Asymmetric groove-side wear was physically observed during inspection and recorded in the field notes. No usable Sheave 1 groove photograph is available because of the camera technical issue.	Direct field observation / ISS inspection records / MEAS-001	Poor	DEF-001	Open
OBS-002	Cable Belt Lower Angle Station Sheave 2	Rope groove / rope path	Top groove depth 106 mm. Front-view photographs CBASSL2-REF001 and CBASSL2-REF002 show a concentrated asymmetric groove-contact band. The wear appears left from the Front and right from the Rear; this is the same physical wear location viewed from opposite perspectives.	CBASSL2-REF001; CBASSL2-REF002	Poor	DEF-002	Open
OBS-003	Cable Belt Lower Angle Station Sheave 3	Rope groove / rope path	Top groove depth 105 mm. Front View photographs CBASSL3-REF001 and CBASSL3-REF002 show the asymmetric groove-wear/contact pattern. The wear appears left from the Front and right from the Rear.	CBASSL3-REF001; CBASSL3-REF002	Poor	DEF-003	Open
OBS-004	Cable Belt Lower Angle Station Sheave 4	Rope groove / rope path	Top groove depth 102 mm. Asymmetric groove-side wear was physically observed during inspection and recorded in the field notes. No usable groove-wear photograph is available because of the camera technical issue. CBASSL4-REF001 and REF002 provide asset and groove-limit traceability.	Direct field observation / CBASSL4-REF001; CBASSL4-REF002; MEAS-004	Poor	DEF-004	Open
OBS-005	Cable Belt Lower Angle Station Sheave 3	Bearing temperature monitoring	Bearing temperature-monitoring sensor observed not attached to the sheave bearing. Temperature trending is not reliable until the sensor is reattached and functionally verified.	Direct field observation	N/A - monitoring deficiency	DEF-003 context	Open
OBS-006	Cable Belt Lower Angle Station Sheave 4	Bearing temperature monitoring	Bearing temperature-monitoring sensor observed not attached to the sheave bearing. Temperature trending is not reliable until the sensor is reattached and functionally verified.	Direct field observation	N/A - monitoring deficiency	DEF-004 context	Open

19 Engineering Assessment

The Angle Station is a coupled rope-path system comprising four corresponding Upper Angle Station sheaves and four Lower Angle Station sheaves. During the inspection, the angular rope transition between the Upper and Lower stations was physically observed. The rope paths are not coincident with the main-sheave groove centre planes, and the same general asymmetric groove-side wear condition was physically identified on all four Lower Angle Station sheaves. On the balance of the observed geometry, the repeated lower-sheave wear pattern and the supporting Upper Sheave No. 4 evidence, the non-zero fleet-angle/off-centre tracking mechanism is assessed as the principal probable cause of the recurring asymmetric Angle Station wear. The outstanding quantitative survey is required to measure the angles/offsets and design the correction, rather than to establish the existence of the angular condition.

The supplied OEM plate identifies a 97 mm nominal rope-groove depth and a 127 mm OEM-defined maximum wear depth, giving a 30 mm vertical wear range. The Lower Sheave readings of 104 mm, 106 mm, 105 mm and 102 mm represent approximately 7 mm, 9 mm, 8 mm and 5 mm of vertical wear respectively, leaving approximately 23 mm, 21 mm, 22 mm and 25 mm before the 127 mm maximum wear-depth limit. These measurements indicate that none of the four Lower sheaves has reached the OEM-defined vertical retirement limit. This does not negate the defect: the principal condition is lateral/asymmetric groove wear, which changes how the rope is supported and can create localized strand compression, pinching, contact stress and fatigue even while the measured depth remains within the vertical wear limit.

The front-view photographs for Lower Sheaves 2 and 3 directly show the localized asymmetric groove-contact/wear pattern. In those images the worn side appears on the left; the same physical wear location appears on the right when viewed from the Rear. This reflects the change in viewing perspective only; the physical wear location is unchanged. Lower Sheaves 1 and 4 were also physically inspected and the same general asymmetric wear condition was observed and recorded, but usable groove photographs were not obtained because of the camera technical issue.

Mechanically, the non-zero fleet angle causes the tensioned rope to seek its natural line of travel and bear against the groove flank instead of remaining continuously supported through the groove centre plane. This produces concentrated rubbing/abrasive contact and is considered a credible mechanism for the accelerated rope and sheave wear. The rope passes through four Lower and four Upper Angle Station sheave interfaces; the angular rope paths between the stations were physically observed, although detailed inspection of Upper Sheaves 1-3 could not be completed because production resumed. Their detailed condition therefore remains to be verified rather than assumed.

The off-centre rope line of action can also alter the sheave load path. A force applied away from the intended sheave centre plane introduces lateral reaction and an overturning/tilting moment that must be reacted by the shaft and bearing pair, potentially creating unequal bearing reactions. The actual bearing loads are not calculated in this report because rope tension, bearing arrangement/spacing and quantitative alignment data were not supplied. Figure 9 is a conceptual load-path illustration only. The bearing temperature-monitoring sensors on Lower Sheaves 3 and 4 were found not attached to the bearings and should be reattached and functionally verified. Available bearing arrangement, vibration and other condition-monitoring history should be reviewed through the mine Condition Monitoring department.

Upper Angle Station Sheave No. 4 is relevant corroborating system evidence. The supplied photographs directly establish collapsed/displaced bearing components, metallic debris and post-failure scoring/smearing of the bearing/hub region, together with asymmetric groove/contact features and an in-situ angular rope/sheave arrangement. These features establish the failure mode and damage pattern and are consistent with the broader concern regarding abnormal off-centre loading. They do not, on their own, constitute a complete root-cause analysis proving fleet angle as the sole cause of that individual bearing collapse.

South32 has advised that the 45 mm cable-belt rope was replaced approximately 20 to 24 months ago and has already experienced wear approaching 50% of its allowable wear allowance, against an expected minimum life of approximately 6 to 8 years. The rope condition has also been independently inspected by another contractor. That contractor's formal report is not included in this report package. Exact replacement dates, historical wear values and the reported 12-to-18-month Angle Station sheave service trend remain mine-controlled information; where verification or further detail is required, the reader should refer the request directly to the mine Condition Monitoring / maintenance functions. Release of the independent contractor report remains at the mine's discretion.

Simply replacing a worn sheave or failed bearing will restore the individual component but will not remove the principal probable geometric driver if the rope continues to enter and leave the Angle Station sheaves at non-zero fleet angles. Owing to the pronounced angular displacement between the Upper and Lower stations and the restricted underground space available for horizontal repositioning of the main sheave support frames, substantial repositioning of the existing main sheaves is not considered a practical primary correction. Engineered guide control at both entry and departure of the affected main sheaves is therefore the recommended corrective solution, subject to

quantitative survey and detailed design verification. CBASS-REF006 to CBASS-REF008 illustrate the functional concept only; they do not prescribe final guide-sheave quantity or configuration. Final positions, diameters, groove profiles, bearings, loads, support reactions, structural steelwork, clearances, guarding, adjustability and maintainability are to be determined by a suitably qualified mechanical/sheave design specialist. The guide function is intended to laterally guide/deflect the rope onto and away from the main-sheave centre plane rather than create a new wrap-around reeving path, and the final design should avoid transferring the same misalignment mechanism to the guide equipment.

20 Recommendations

The recommendations below link the identified findings to corrective actions, responsibilities, verification requirements and close-out evidence.

Table 20 - Detailed recommendation and action register

Action ID	Related defect ID	Asset	Recommendation	Priority	Responsible discipline	Target / shutdown	Verification requirement	Close-out evidence	Status
REC-001	DEF-001 to DEF-004	Full Upper/Lower Angle Station	Complete 8-sheave geometry survey and inspect Upper Sheaves 1-3 during design discovery. See Section 20.1.	P3	Mechanical / Engineering	Short term	Signed as-found geometry	Survey + Upper inspection record	Open
REC-002	DEF-001 to DEF-004	Full Upper/Lower Angle Station	Engineer entry/departure guide control; target 0 degrees. Specialist to set final guide quantity/configuration. Figures are contextual only. See Section 20.1.	P3	Mechanical / Engineering	Design short term; planned shutdown	Approved design + as-left tracking	Design / installation record	Open
REC-003	DEF-001 to DEF-004	Lower sheaves / bearings	Reattach Sheaves 3/4 temperature sensors and continue mine vibration/temperature plus groove/rope monitoring. Escalate readings outside mine limits.	P3	Mechanical / Condition Monitoring	Immediate sensors; ongoing	Credible sensors + CM review	Sensor / CM records	Open
REC-004	DEF-001 to DEF-004	Lower main sheaves / Upper counterparts	Replace the worn Lower main sheave when its guide correction is commissioned. Replace corresponding Upper sheave if condition requires; replace earlier if monitoring deteriorates.	P3	Mechanical / Maintenance	Modification shutdown or earlier	New Lower sheave + Upper disposition	Replacement records	Open
REC-005	DEF-001 to DEF-004	Modified rope paths	Verify as-left geometry and centred running. Trend groove, rope and bearing condition against baseline at 6-12 months and longer term.	P3	Mechanical / Inspection / CM	Commissioning + follow-up	Geometry + trend comparison	Commissioning / trend record	Open
REC-006	System context / history	Cable Belt / Angle Station history	Confirm mine-controlled rope/sheave history. Reference third-party rope report only if the mine releases it; direct verification requests to the mine.	P3	Reliability / Maintenance Planning	Short term	Mine-confirmed baseline	CMMS / mine confirmation	Open

20.1 Recommended Corrective Implementation Strategy

The recommended physical correction is an engineered guide-sheave system that provides rope-path control at both the entry and departure of every affected main Angle Station sheave. The functional requirement is that the final rope segment approaches the main sheave, passes through the groove and departs on the intended groove centre plane at 0 degrees fleet angle, or as close as reasonably practicable. The guide-sheave drawings in Section 14 illustrate this functional objective only. They are not design or construction guidelines, and the final number, position and arrangement of guide sheaves are to be established by a suitably qualified mechanical/sheave design specialist from the quantitative survey and design-discovery information.

The outstanding detailed inspection of Upper Angle Station Sheaves 1-3 should be completed during the same survey/design-discovery phase rather than being treated as a separate prerequisite that delays commencement of the design work. The design phase is to establish the final guide-sheave size, groove compatibility, bearing duty, support loads, structural attachment, adjustment, guarding, access and maintenance requirements. The intended guide function is lateral guidance/deflection of the rope into the required centre-line condition; the final arrangement should not introduce a new wrap-around reeving path, unacceptable additional bending or a new fleet-angle/misalignment condition at the guide equipment.

The existing Lower Angle Station main sheaves may remain in short-term service under the P3 planned-maintenance framework, subject to the mine's operational risk controls and enhanced condition monitoring. They are not recommended as the long-term main sheaves for the corrected geometry because the groove profiles are already asymmetrically distorted and may continue to damage or steer the rope. Accordingly, when a rope-path guide correction is commissioned, the corresponding existing worn Lower main sheave should be replaced with a new sheave during the same maintenance intervention. This avoids exposing a new sheave to the existing misalignment mechanism and avoids commissioning the corrected rope path through an already damaged Lower groove. The corresponding Upper main sheave condition is to be assessed during design discovery and replacement incorporated where the groove or bearing condition is not suitable for continued service. If the mine's condition monitoring identifies bearing or other deterioration before the planned modification, the affected sheave should be replaced at the earliest practicable maintenance opportunity.

The mine may implement the correction across all four rope paths as one planned programme, which provides the most direct system-wide correction, or may elect to use a controlled staged proof-of-concept. If a proof-of-concept is selected, mine maintenance history should be used to identify a representative or historically problematic rope path; for example, the No. 4 rope path may be considered if the mine records support that selection. The pilot should implement the complete functional solution on that rope path - guide control at both entry and departure of the corresponding Upper and Lower main sheaves, together with replacement of the worn Lower main sheave and any Upper main-sheave replacement indicated by condition. The modified rope path should then be trended for approximately 6 to 12 months before wider rollout. A staged trial does not defer earlier intervention on any other sheave that develops condition-monitoring indications outside the mine's defined limits.

Before and after the modification, repeatable groove, rope and condition-monitoring baselines should be retained. Following installation, the as-left geometry and fleet angles should be recorded and the rope should be observed under operating conditions to verify stable centred tracking through the main groove. Initial success should be assessed by improvement in groove wear rate, rope wear rate and bearing vibration/temperature trends relative to the current condition. Longer-term effectiveness should be assessed as sufficient operating history accumulates by comparing achieved rope and sheave service life with the historical performance; no fixed long-term service-life target is set by this report.

21 Conclusion

Within the inspection scope, all four Cable Belt Lower Angle Station sheaves are recorded as Poor condition with Level 2 - Moderate severity and P3 - Planned maintenance priority. All four were physically inspected and the same general asymmetric groove-side wear condition was identified. The vertical groove-depth readings of 102-106 mm remain inside the OEM-defined 127 mm maximum wear-depth limit, but the asymmetric sidewall profile means the grooves are no longer providing optimum centred support to the 45 mm rope. The angular Upper-to-Lower rope transition was physically observed, and the non-zero fleet-angle/off-centre tracking condition is assessed as the principal probable cause of the recurring asymmetric wear. Quantitative survey remains necessary to measure the actual geometry and to finalize corrective design dimensions.

South32 has advised that the rope has experienced materially shorter life than expected and that Angle Station sheaves have experienced repeat short service intervals. The rope wear condition has also been independently inspected by another contractor; its formal report is not included in this package and remains mine-controlled information. Appendix F directly documents the Upper Angle Station Sheave No. 4 collapsed-bearing failure, metallic debris, post-failure scoring/smearing and asymmetric groove/contact condition. This evidence corroborates the wider

system concern and failure pattern, while the Upper Sheave No. 4 event remains within the broader reliability/condition-monitoring review rather than being assigned a separate causal conclusion in this report. Detailed condition of Upper Sheaves 1-3 remains to be verified during the survey/design-discovery phase because their planned inspection was curtailed when the plant returned to production.

Replacing worn sheaves or failed bearings without changing the rope path will not remove the principal probable geometric driver. The recommended corrective solution, subject to detailed design verification, is engineered guide control at both rope entry and rope departure so that each main sheave is traversed on its intended centre plane at 0 degrees fleet angle, or as close as reasonably practicable. The conceptual drawings define the required function only; final guide-sheave quantity and configuration are to be established by specialist design. For each rope path when the correction is commissioned, the existing worn Lower main sheave should be replaced with a new sheave in the same intervention, unless condition monitoring requires earlier replacement. The mine may elect a full four-path rollout or a controlled proof-of-concept followed by approximately 6 to 12 months of trend verification. Bearing temperature sensors on Lower Sheaves 3 and 4 should be reattached and verified, and interim vibration/temperature and other mine condition-monitoring controls should remain active until the corrective programme is completed. Post-correction effectiveness should be evaluated by as-left geometry and centred tracking verification, followed by groove wear, rope wear, bearing vibration/temperature and service-life trends.

22 Technical Review and Report Issue Sign-off

This sign-off confirms technical review and issue of the report; it does not transfer the client's responsibility for site risk assessment, operational decisions or implementation of corrective actions.

Table 21 - Technical review and issue sign-off

Role	Name	Title / discipline	Review status	Date	Signature
Prepared by	Brendan Wood	Projects / Systems	Report preparation	11/08/2026	
Technical reviewer	Kevin Wood	Managing Engineer	Review Complete.	17/08/2026	
Approved for issue	Kevin Wood	Managing Engineer	Approval	17/08/2026	

SECTION D

APPENDICES &
EVIDENCE

Photographic records, checklists, measurements, NDT,
drawings and supporting documentation.



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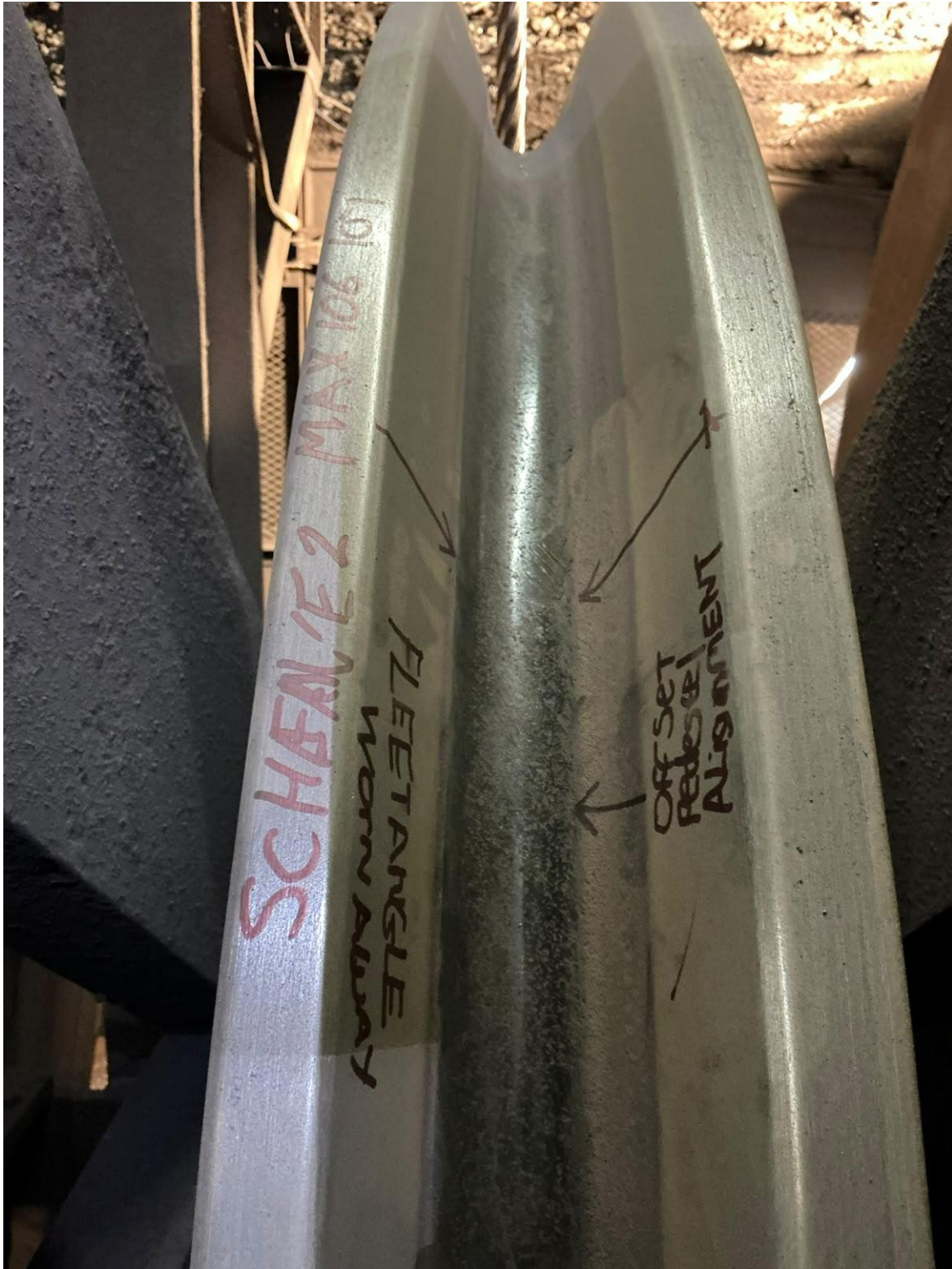
Appendix A - Photographic Register

Photographic Register

Photo ID	Asset	Description / observation	Related defect ID	File name / source	Status
CBASSL2-REF001	Cable Belt Lower Angle Station Sheave 2	Front-view contextual groove photograph showing the asymmetric polished/contact band and localized groove-side wear on Lower Angle Station Sheave 2.	DEF-002	CBASSL2-REF001-Wear Pattern.jpeg	Included in Appendix A
CBASSL2-REF002	Cable Belt Lower Angle Station Sheave 2	Front-view close/contextual groove photograph showing asymmetric polished/contact wear along the groove sidewall/base on Lower Angle Station Sheave 2.	DEF-002	CBASSL2-REF002-Wear Pattern.jpeg	Included in Appendix A
CBASSL3-REF001	Cable Belt Lower Angle Station Sheave 3	Front-view overview of Lower Angle Station Sheave 3 showing the asymmetric groove-wear/contact pattern and marked inspection context.	DEF-003	CBASSL3-REF001-Wear Pattern Taken from the front view.jpeg	Included in Appendix A
CBASSL3-REF002	Cable Belt Lower Angle Station Sheave 3	Front-view close-up of CBASSL3-REF001 showing localized asymmetric groove wear/contact on Lower Angle Station Sheave 3.	DEF-003	CBASSL3-REF002-Wear Pattern Close-up of CBASSL3-REF001.jpeg	Included in Appendix A
CBASSL4-REF001	Cable Belt Lower Angle Station Sheave 4	OEM/asset plate showing Kimberley Engineering Works manufacturer information, 1900 mm sheave diameter, 45 mm rope diameter and date of manufacture 01-2022.	DEF-004	CBASSL4-REF001-Sheave Attributes.jpeg	Included in Appendix A

Photo ID	Asset	Description / observation	Related defect ID	File name / source	Status
CBASSL4-REF002	Cable Belt Lower Angle Station Sheave 4	OEM groove-depth limit plate recording A = Rope Groove Depth 97 mm and B = Max Wear Depth 127 mm.	DEF-004	CBASSL4-REF002- Limit Specification.jpeg	Included in Appendix A

Photographic Evidence Pages



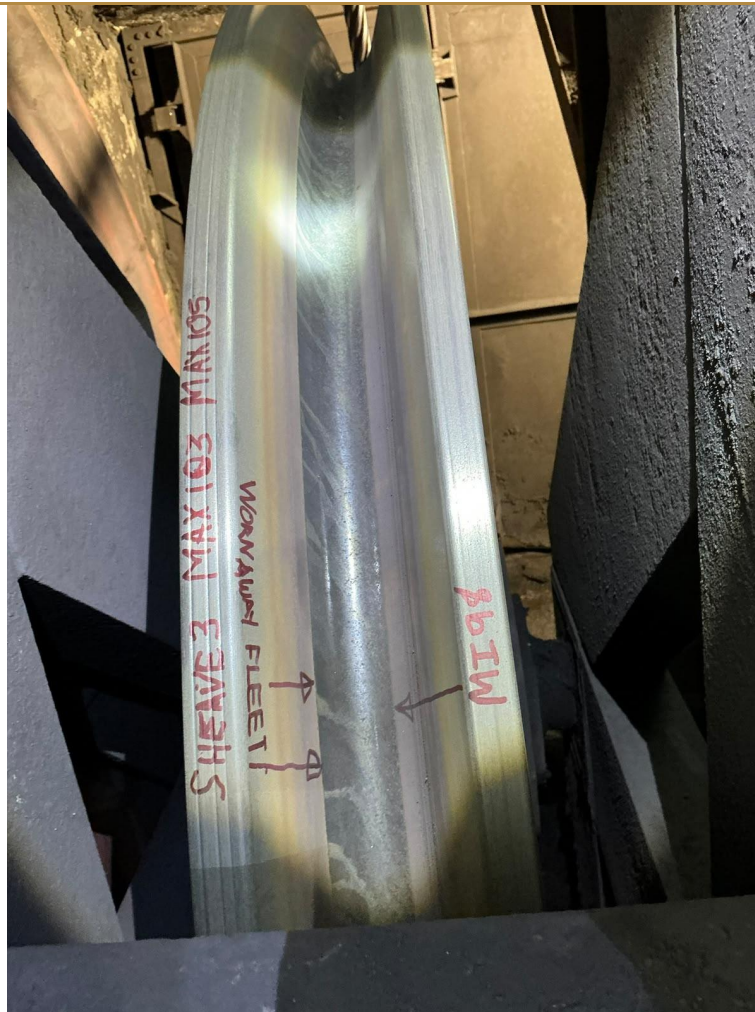
CBASSL2-REF001 - Cable Belt Lower Angle Station Sheave 2

Front-view contextual groove photograph showing the asymmetric polished/contact band and localized groove-side wear on Lower Angle Station Sheave 2. The wear appears left in Front View and right in Rear View.



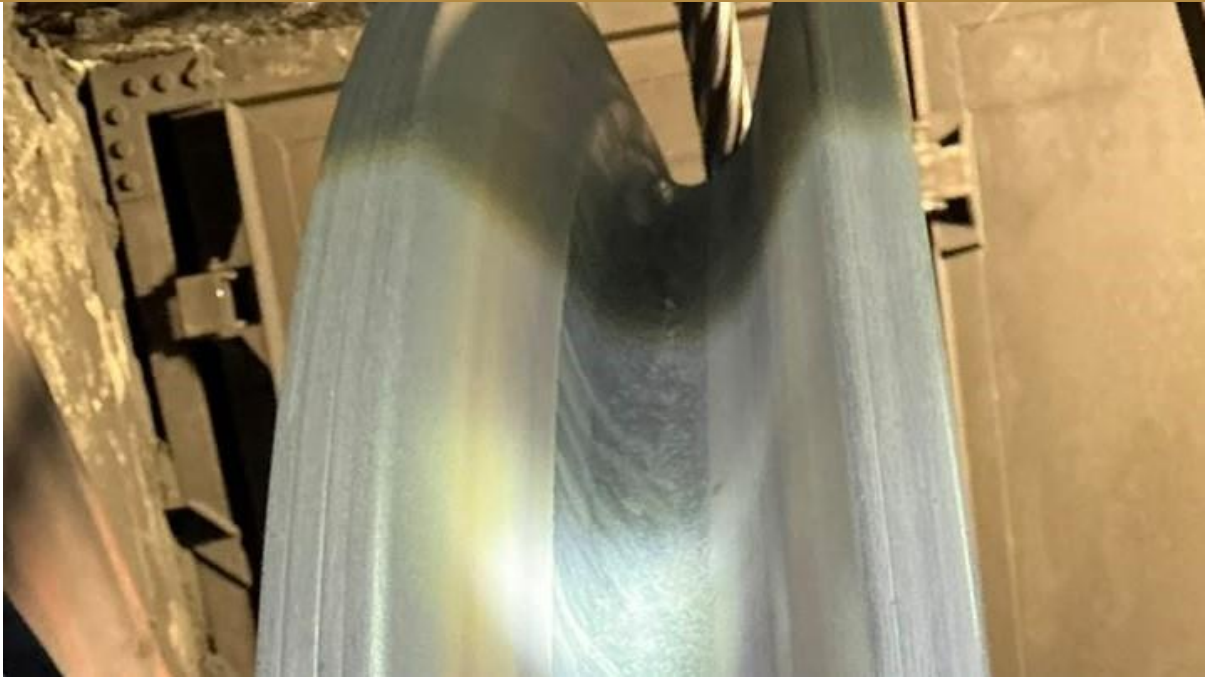
CBASSL2-REF002 - Cable Belt Lower Angle Station Sheave 2

Front-view close/contextual groove photograph showing asymmetric polished/contact wear along the groove sidewall/base on Lower Angle Station Sheave 2. The wear appears left in Front View and right in Rear View.



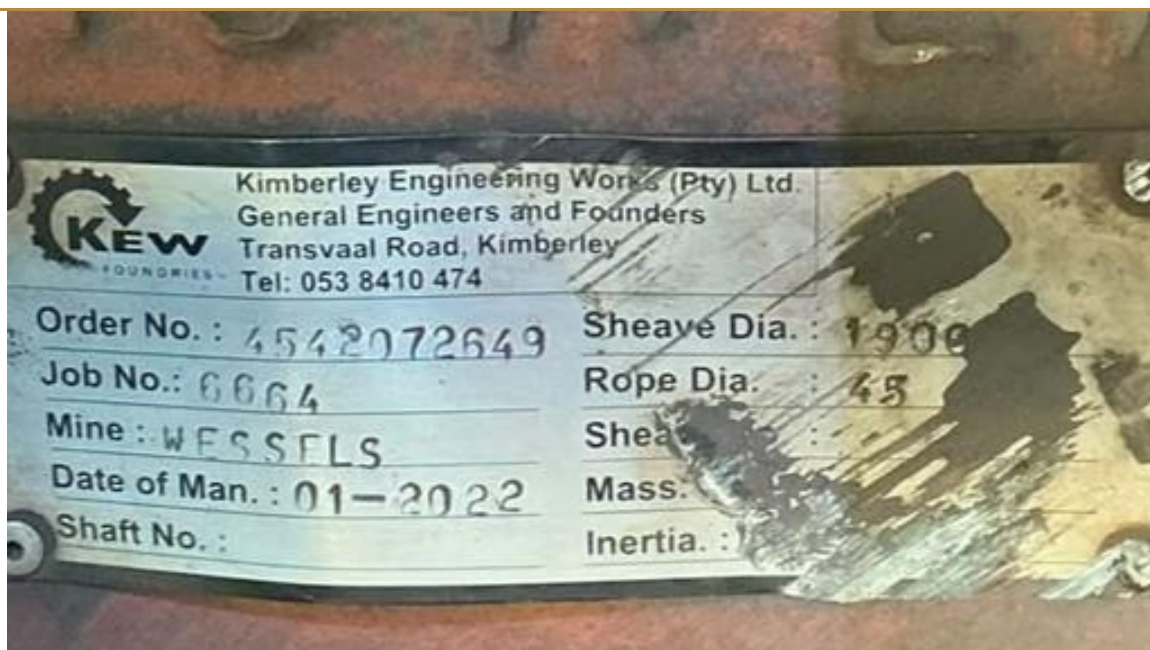
CBASSL3-REF001 - Cable Belt Lower Angle Station Sheave 3

Front-view overview of Lower Angle Station Sheave 3 showing the asymmetric groove-wear/contact pattern and marked inspection context. The wear appears left in Front View and right in Rear View.



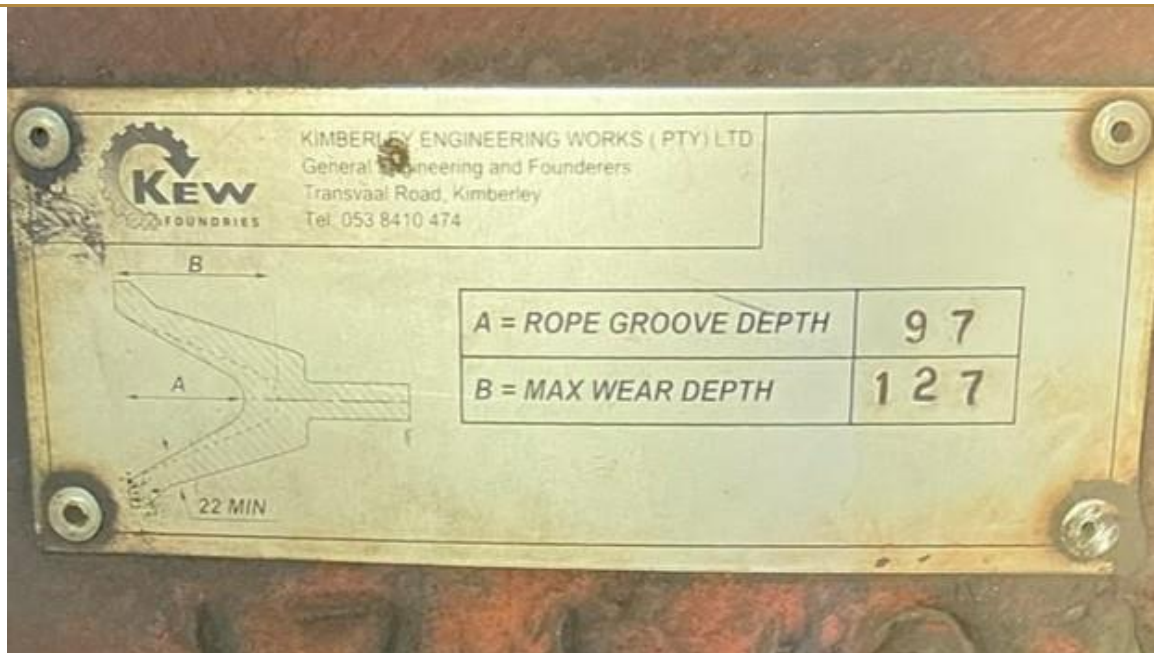
CBASSL3-REF002 - Cable Belt Lower Angle Station Sheave 3

Front-view close-up of CBASSL3-REF001 showing the localized asymmetric groove-wear/contact pattern on Lower Angle Station Sheave 3.



CBASSL4-REF001 - Cable Belt Lower Angle Station Sheave 4

OEM/asset plate showing Kimberley Engineering Works manufacturer information, 1900 mm sheave diameter, 45 mm rope diameter and date of manufacture 01-2022.



CBASSL4-REF002 - Cable Belt Lower Angle Station Sheave 4

OEM groove-depth limit plate recording A = Rope Groove Depth 97 mm and B = Max Wear Depth 127 mm.

Appendix B - Inspection Checklist

Table 22 - Angle Station sheave inspection checklist

Item	Inspection check	Result	Observation / limitation	Evidence reference	Related defect ID
1	Asset identification confirmed	Partial	Sheave positions 1–4 and available OEM/manufacture information were recorded; individual asset tags were not supplied.	Sections 9 and 15; CBASSL4-REF001	N/A
2	Rope groove and rim condition examined	Completed	All four Lower sheave grooves were visually inspected and groove depths measured. Asymmetric groove-side wear was identified on all four; photographic coverage is limited for Sheaves 1 and 4.	Sections 16.1–16.4; MEAS-001 to MEAS-004; CBASSL2-REF001/002; CBASSL3-REF001/002	DEF-001 to DEF-004
3	Rope tracking and alignment indicators examined	Completed – qualitative	The angular Upper-to-Lower rope transition and off-centre tracking condition were observed. Quantitative fleet-angle/alignment measurements were not recorded.	Sections 16.1–16.5; CBASS-REF005 to CBASS-REF008	DEF-001 to DEF-004
4	Hub, shaft and bearing housing externally examined	Limited	Assessment was limited to accessible external condition; internal bearing condition was not established. Bearing temperature sensors on Sheaves 3 and 4 were found detached.	Sections 16.3–16.5; OBS-005; OBS-006	DEF-003 / DEF-004 context
5	Structural supports, fasteners and guards examined	Limited	Visible surrounding structures were reviewed contextually; no dedicated structural defect or NDT finding was assigned.	Sections 16.1–16.4	N/A
6	Corrosion, coating and housekeeping assessed	Completed – visual	No material corrosion/coating defect was assigned from the available evidence. Local surface contamination/operating residue was observed.	Sections 16.1–16.4	N/A

Appendix C - Measurements and Dimensional Records

Table 23 - Measurements and dimensional records

Measurement ID	Asset	Location / feature	Instrument / equipment ID	Measured value	Unit	Reference / criterion	Deviation	Comment
MEAS-001	Cable Belt Lower Angle Station Sheave 1	Top rope groove depth	009741746	104	mm	Supplied reference: 97 mm rope-groove depth / 127 mm maximum wear depth	Acceptance deviation not assessed	Recorded in ISS inspection records; Bosch PLR 50 calibration certificate not supplied.
MEAS-002	Cable Belt Lower Angle Station Sheave 2	Top rope groove depth	009741746	106	mm	Supplied reference: 97 mm rope-groove depth / 127 mm maximum wear depth	Acceptance deviation not assessed	Recorded in ISS inspection records; Bosch PLR 50 calibration certificate not supplied.
MEAS-003	Cable Belt Lower Angle Station Sheave 3	Top rope groove depth	009741746	105	mm	Supplied reference: 97 mm rope-groove depth / 127 mm maximum wear depth	Acceptance deviation not assessed	Recorded in ISS inspection records; Bosch PLR 50 calibration certificate not supplied.
MEAS-004	Cable Belt Lower Angle Station Sheave 4	Top rope groove depth	009741746	102	mm	Supplied reference: 97 mm rope-groove depth / 127 mm maximum wear depth	Acceptance deviation not assessed	Recorded in ISS inspection records; Bosch PLR 50 calibration certificate not supplied.

Appendix D - NDT Results

Table 24 - NDT results register

NDT ID	Asset	Location	Method	Procedure / technique	Operator / certification	Result	Report reference	Comment
N/A	Lower Angle Station Sheaves	N/A	No NDT evidence supplied	N/A	N/A	Not assessed	Not supplied	No NDT results were included in the supplied package.

Appendix E - Drawings, Schematics and Reference Information

Drawings and Schematics Register

Reference ID	Drawing / document number	Title	Revision	Source	Status	Comment
CBASS-REF001	CBASS-REF001	Normal Wear Pattern	1.0	Supplied report package	Reference only	Normal centred groove-contact context.
CBASS-REF002	CBASS-REF002	Mis-aligned Groove Wear - Left Hand Side	1.0	Supplied report package	Reference only	Abnormal left-side groove-wear context.
CBASS-REF003	CBASS-REF003	Mis-aligned Groove Wear - Right Hand Side	1.0	Supplied report package	Reference only	Abnormal right-side groove-wear context.
CBASS-REF004	CBASS-REF004	Front-Rear Reference	1.0	Supplied report package	Reference only	Defines front/rear orientation relative to cable belt.
CBASS-REF005	CBASS-REF005	Cable Belt Lower Sheaves Top View	1.0	Supplied report package	Reference only	Contextual rope-path/fleet-angle offset.
CBASS-REF006	CBASS-REF006	Angle Station Lower Sheaves Rear View	1.0	Supplied report package	Reference only	Recommended correction concept - rear view; functional context only, not final guide quantity/configuration.
CBASS-REF007	CBASS-REF007	Angle Station Lower Sheaves Top View	1.0	Supplied report package	Reference only	Recommended correction concept - top view; functional context only, not a construction/design guideline.
CBASS-REF008	CBASS-REF008	Side View of Proposed Guide Sheaves	1.0	Supplied report package	Reference only	Conceptual side view of lateral guide control; final geometry subject to specialist detailed design.
CBASS-REF009	CBASS-REF009	Angular Forces on Sheave Bearings	1.0	Supplied report package	Reference only	Conceptual bearing-load effect of rope offset.

Appendix F - Supporting Documentation

Table 25 - Supporting documentation register

Document ID	Title / description	Document number	Revision / date	Source	Report relevance	Status
DOC-001	Layout Reference Images	Layout Reference Images.zip	Supplied image set	Project evidence package	Engineering reference drawings CBASS-REF001 to REF009	Included

Document ID	Title / description	Document number	Revision / date	Source	Report relevance	Status
DOC-002	Lower Sheave photographic evidence	Sheave 2 / Sheave 3 / Sheave 4 image ZIP packages	Supplied image sets	Project evidence package	Inspection photographs and asset/specification plate evidence	Included
DOC-003	Upper Angle Station Sheave No. 4 - Set One post-failure / post-extraction photographs	Upper Angle Station Sheave No 4 Set One.zip	July 2026	Project evidence package	Contextual supporting evidence: groove/contact wear, post-extraction sheave/hub condition and OEM plates	Included in Appendix F
DOC-004	Upper Angle Station Sheave No. 4 - Set Two failure-mode photographs	Upper Angle Station Sheave No 4 Set Two.zip	15/07/2026 photographs	Project evidence package	Contextual supporting evidence: collapsed bearing, angular rope/sheave arrangement and metallic debris	Included in Appendix F

F.1 Upper Angle Station Sheave No. 4 Failure - Contextual Supporting Evidence

The following photographs are included to provide system-level context for the reported Upper Angle Station Sheave No. 4 failure. South32 notes identify the failure date as 14 July 2026 and the failure mode as collapsed bearing. Set Two photographs carry 15 July 2026 camera timestamps and document the in-situ failed bearing condition. Set One contains later post-extraction/post-inspection photographs of the removed sheave, groove/contact condition, hub/bearing-seat areas and plates. These images are supporting reference only; this Lower Angle Station report does not assign a separate Upper Angle Station defect classification or claim that fleet angle was the sole root cause of the bearing failure.

F.1.1 Set One

Photographs are retained under their supplied file/reference names for traceability.



CBASSUF4-REF001-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction view of the removed Upper Angle Station Sheave No. 4 groove with contextual scale/marking.



CBASSUF4-REF002-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction groove view showing localized asymmetric contact/wear on the removed Upper Angle Station Sheave No. 4.



CBASSUF4-REF003-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction groove view showing localized asymmetric contact/wear on the removed Upper Angle Station Sheave No. 4.



CBASSUF4-REF004-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction groove view showing localized asymmetric contact/wear on the removed Upper Angle Station Sheave No. 4.



CBASSUF4-REF005-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction groove view showing localized asymmetric contact/wear on the removed Upper Angle Station Sheave No. 4.



CBASSUF4-REF006-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction groove view showing localized asymmetric contact/wear on the removed Upper Angle Station Sheave No. 4.



CBASSUF4-REF007-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction view of the removed Upper Angle Station Sheave No. 4 groove with contextual scale/marking.



CBASSUF4-REF008-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction overview of the Upper Angle Station Sheave No. 4 hub/bearing region.



CBASSUF4-REF009-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction close-up of the hub/bearing-seat region showing severe scoring/damage associated with the failed assembly.



CBASSUF4-REF010-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction close-up of the hub/bearing-seat region showing severe scoring/damage associated with the failed assembly.



CBASSUF4-REF011-Extracted Failed upper angle station Sheave 4 post inspection.jpg - post-extraction close-up of the hub/bearing-seat region showing severe scoring/damage associated with the failed assembly.



CBASSUF4-REF012-Extracted Failed upper angle station Sheave 4 post inspection.jpg - Post-extraction OEM/asset plate and groove/asset specification reference for Upper Angle Station Sheave No. 4.



CBASSUF4-REF013-Extracted Failed upper angle station Sheave 4 post inspection.jpg - Post-extraction OEM/asset plate and groove/asset specification reference for Upper Angle Station Sheave No. 4.



CBASSUF4-REF014-Extracted Failed upper angle station Sheave 4 post inspection.jpg - Post-extraction OEM/asset plate and groove/asset specification reference for Upper Angle Station Sheave No. 4.

F.1.2 Set Two

Photographs are retained under their supplied file/reference names for traceability.



CBASSU4-REF001-Failure Mode - Bearing.jpeg - In-situ view of Upper Angle Station Sheave No. 4 showing severe collapsed/displaced bearing components.



CBASSU4-REF002-Failure Mode - Bearing.jpeg - Close in-situ view into the failed bearing/housing showing displaced and damaged internal bearing components.



CBASSU4-REF003-Failure Mode - Angular alignment.jpeg - In-situ side view showing the rope approaching the Upper Angle Station sheave at an angular offset relative to the sheave plane.



CBASSU4-REF004-Failure Mode - Bearing.jpeg - Additional in-situ view of the collapsed/displaced bearing assembly at Upper Angle Station Sheave No. 4.



CBASSU4-REF005-Failure Mode - Bearing Shavings.jpeg - Metallic bearing debris/shavings recorded in the vicinity of the failed Upper Angle Station Sheave No. 4 assembly.